# Do Reasoning-Quality Metrics Predict Better Decisions? Verifying Multi-Agent Debate Against External Market Outcomes

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## Abstract

Multi-agent LLM systems increasingly use structured debate to improve reasoning, and their outputs are commonly evaluated with internal reasoning-quality rubrics and consensus measures. Whether those proxies predict better *decisions* is rarely tested, because most benchmarks lack an external outcome to check them against. We use simulated portfolio allocation as a testbed where complete reasoning traces can be compared against an objective downstream outcome, evaluating specialist LLM agents under single-agent, independent-ensemble, and multi-round debate conditions, scored with a CRIT-style four-pillar rubric and Jensen–Shannon portfolio divergence against realized return, Sharpe, and drawdown. Across 210 runs, CRIT reasoning scores are essentially uncorrelated with financial outcomes (descriptive $r{=}+0.07$ with Sharpe, $r{=}+0.03$ with return), even though structured prompting moves those scores substantially ($\bar{\rho}$ from $0.72$ to $0.84$, Cohen's $d{\approx}2.0$) and debate portfolios beat an equal-weight baseline only 33% of the time. Improved outcomes align with interventions that limit consensus collapse rather than with interventions that raise reasoning scores: a divergence-collapse trigger improves Sharpe by $+0.14$ ($p{=}0.028$), while an intervention that verifiably repairs causal-reasoning scores slightly degrades performance. That trigger also constrained revision magnitude, so we cannot isolate disagreement preservation as the mechanism. Across our tested configurations the clearest benefits concern risk and risk-adjusted performance rather than consistent alpha, and consensus and rubric scores can both be maximized while decision quality stagnates. These results suggest internal reasoning proxies should be validated against external task outcomes before being treated as evidence of agent decision quality.

## 1 Introduction

Multi-agent LLM systems increasingly resolve hard problems by deliberation: agents propose, critique, and revise before committing to an answer. Because the ground truth for such tasks is often unavailable at evaluation time, these systems are routinely judged by *internal proxies*—rubric-based reasoning scores, self-consistency, inter-agent agreement—and improvements in those proxies are treated as evidence that the system is deciding better.

That inference is rarely tested. A rubric can reward an argument that is coherent, well-cited, and causally explicit without any guarantee that the resulting *decision* is better; consensus can reflect social capitulation as easily as error correction. Establishing whether a proxy tracks decision quality requires a task with an external, objectively measurable outcome that the evaluator cannot see. Most reasoning benchmarks do not provide one.

We therefore study the proxy–outcome relationship directly, in a setting that does. Simulated portfolio allocation supplies conflicting evidence, complete auditable reasoning traces, and an objective

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downstream outcome (realized return, Sharpe ratio, and drawdown over a held-out quarter) measured strictly after the decision is made. We stress that finance is the *testbed*, not the subject: we make no claim that these agents are deployable traders, and realized return is a noisy outcome signal rather than ground-truth reasoning correctness (Section 5). We ask three questions:

**RQ1. Do reasoning-quality metrics predict financial performance?** If a CRIT-style rubric captures decision-relevant reasoning, higher scores should associate with better outcomes. *Falsifier:* near-zero association, or rubric gains alongside flat or worse performance.

**RQ2. Does structured debate improve reasoning and portfolio performance?** Debate may expose weak arguments and improve decisions. *Competing explanation:* any gain comes from having multiple perspectives at all, not from interaction. *Falsifier:* an independent ensemble with no communication matches or beats debate on matched scenarios.

**RQ3. Can targeted interventions repair reasoning failures during debate?** *Falsifier:* interventions that verifiably repair the measured reasoning deficit leave outcomes unchanged or worse.

Our results answer all three, and the answers diverge from what the proxies alone would suggest. Reasoning scores are highly controllable—structured prompting raises mean CRIT quality $\bar{\rho}$ from $0.72$ to $0.84$—yet across 210 repeated-condition runs $\bar{\rho}$ is essentially uncorrelated with Sharpe (descriptive $r{=}{+}0.07$) and with return ($r{=}{+}0.03$), and debate portfolios beat an equal-weight $1/N$ baseline only 33% of the time. Debate's measurable benefit is risk reduction rather than alpha. The only intervention producing a significant risk-adjusted gain is one that blocks premature convergence ($+0.14$ Sharpe, $p{=}0.028$; raw return $+0.83$pp, $p{=}0.052$), whereas one that verifiably repairs causal-reasoning scores slightly degrades performance.

**Contributions.**

(1) We provide evidence that a four-pillar reasoning rubric can be moved substantially by prompt design (Cohen's $d{\approx}2.0$) while remaining uninformative about the external outcome it is used to justify: a proxy-validity failure measured against a decision outcome rather than against another rubric.

(2) We evaluate debate against matched single-agent and *independent ensemble* controls on identical scenarios, separating the effect of interaction from the effect of agent diversity, a control frequently missing when debate is credited with a gain.

(3) We identify and operationalize *consensus convergence*: agents move toward the group position during revision without introducing new evidence, with portfolio divergence collapsing after the first revision and no accompanying outcome gain. This coincides with loss of ensemble diversity and no outcome gain.

(4) We show reasoning quality and decision quality can be regulated independently: an intervention that limits consensus collapse improves risk-adjusted outcomes while one that repairs the weakest reasoning pillar does not, though the former also constrains revision size and so does not isolate the mechanism.

### 1.1 Related Work

**Multi-agent debate.** Du et al. [2023] show that having model instances propose and challenge each other over rounds improves reasoning and reduces hallucination relative to single-shot generation. That result, and the systems built on it, are evaluated on tasks with known answers; our question is what happens when the only available check is a rubric.

**LLM agents in markets.** Xiao et al. [2024] model a trading firm whose specialized agents debate before trading, reporting improved metrics over simple baselines; Li et al. [2025] emphasize evaluation protocols beyond profitability; Zhao et al. [2025] evaluate role-based collaboration on stock-picking under varying risk tolerance; and Lopez-Lira [2025] use market simulations to test financial theories with LLM agents. These works largely report whether a multi-agent configuration performs better. None, to our knowledge, tests whether the reasoning scores used to justify such systems track the outcome. Simulators such as ABIDES [Byrd et al., 2019, Amrouni et al., 2021] supply microstructure realism but expose little agent reasoning, and production systems prioritize robustness over auditability [FTI Consulting, 2023].

**Reasoning evaluation and control.** RAudit [Chang and Geng, 2026] scores reasoning as the mean $\rho$ of four pillars (logical validity $P_1$, evidential support $P_2$, alternative consideration $P_3$, causal alignment $P_4$), measures agreement via Jensen–Shannon divergence, and shows $\rho$ can be stabilized

by PID control without ground-truth access. Chang [2026] target trace–output consistency to reduce sycophancy via retries with escalating feedback. Both establish that reasoning quality is *measurable and controllable*; neither establishes that moving it changes external decision quality, which is the gap we address.

## 2 System and Evaluation Setup

Our framework has four modules: data generation, market simulation, multi-agent debate, and evaluation. The design requirement that matters here is that reasoning be simultaneously *producible*, *measurable*, and *regulable*, so the proxy–outcome relationship can be examined without the measurement contaminating the decision. Each subsystem sees the others only through defined interfaces, and each round writes structured artifacts for proposals, critiques, revisions, controller state, and portfolio decisions. The financial analyses reported below are regenerated from these stored artifacts rather than from runtime state; the per-round CRIT records underlying RQ1 were not preserved in the released artifact set, which we disclose in Section 5. Scenario memos are distilled from FRED macro data, SEC EDGAR filings [U.S. Securities and Exchange Commission, 2024], Yahoo Finance prices [Yahoo Finance, 2024], FinnHub news sentiment [Finnhub, 2024], and earnings transcripts; each quantitative datum carries a stable evidence ID (e.g. `[L1-FF]`) that agents must cite to defend a claim (Appendix C.23). Each run initializes a fresh cash portfolio over a universe of up to 35 tickers, and a broker component logs the trades implied by the agreed allocation.

**Debate protocol.** The orchestrator is a LangGraph `StateGraph` [LangChain, Inc., 2024] running `load_scenario` $\rightarrow$ `propose` $\rightleftharpoons$ `critique` $\rightleftharpoons$ `revise` $\rightarrow$ `judge`. For each round $t = 1, \ldots, T_{\max}$ agents (1) propose an allocation over the scenario's ticker universe, (2) critique other agents' proposals, and (3) revise their own in light of those critiques. Six specialist roles are implemented (macro, value, risk, technical, sentiment, and a devil's advocate configured to disagree), though experiments use subsets of 3–4. The debate is aggregated into one portfolio either by an LLM judge that synthesizes final revisions with their justifications, or by an unweighted mean of each agent's last revision; the latter is our *independent-ensemble* control when critique and revision are disabled.

**Reasoning measurement and control.** Following an RAudit-style scheme [Chang and Geng, 2026], a CRIT evaluator (Critical Reading Inquisitive Template) [Chang, 2023] scores each agent's reasoning on four pillars $P_1^{(i)}, \ldots, P_4^{(i)} \in [0, 1]$, averaged into a per-agent score $\rho_i = \frac{1}{4}\sum_k P_k^{(i)}$ and a group mean $\bar{\rho} = \frac{1}{n}\sum_i \rho_i$. A PID controller drives $\bar{\rho}$ toward a target $\rho^\star$ by regulating a contentiousness parameter $\beta_t \in [0, 1]$ that sets critique tone. Debate ends at $T_{\max}$ or on PID early stop (high reasoning quality with low diversity: $\mathrm{JSD}_t < \epsilon$ and $|\bar{\rho}_t - \bar{\rho}_{t-1}| < \Delta\rho$). Crucially, the evaluator never sees realized prices or any post-decision outcome, so $\bar{\rho}$ and the financial metrics are measured independently. A separate consistency evaluator implementing Regulated Causal Anchoring (RCA) [Chang, 2026] checks whether an agent's revision rationally addresses the critiques it received or merely capitulates; when it fails, the turn is retried with feedback whose tone is set by the PID state.

## 3 Results

### 3.1 Experimental Setup

We evaluate agent allocations through backtesting on real historical market data. Agents construct portfolio allocations on the first day of each quarter using the preceding quarter's data, hold positions through the quarter, and are evaluated daily. Our largest-scale experiments use 35 market scenarios spanning 12 quarters (2021Q4–2025Q3), covering multiple regimes including the 2022 bear market, the regional banking crisis, and the AI-driven rally. Ticker universes range from 3–35 tickers (mean 11.5) across seven sectors; cross-scenario similarity is low (mean Jaccard 0.216), with intentional skew toward 2022 as a stress-test for reasoning interventions. Daily prices from Yahoo Finance plus S&P 500 (SPY) enable computation of daily equity curves and meaningful financial metrics beyond return: Sharpe and Sortino ratios, volatility, and maximum drawdown. Unless otherwise specified, agents use `gpt-5-mini` (temperature 0.3); CRIT evaluation uses `gpt-5`. Agent behavior is parameterized through three configuration layers: a **debate configuration** (orchestration protocol, agent roles, rounds, LLM provider, PID gains); a **scenario configuration** (quarter, ticker universe, sector mappings, allocation constraints); and per-role **agent profiles** (YAML files declaring ordered prompt blocks, structured user-prompt templates, and required variables for each debate phase).

**Reasoning profile tiers.** Agent prompts exist at three tiers of increasing structure, which is the independent variable in our RQ2 prompt experiments. **Baseline** profiles (Appendix C.20) give a minimal role definition (∼30 lines) plus a contract requiring bracketed evidence citation. **Causal scaffolding** profiles (Appendix C.21) expand this to ∼140 lines, adding disagreement tracking, epistemic calibration, weakest-assumption attacks, a narrative commitment (e.g. "macroeconomic regimes drive sector leadership"), and differentiation rules that keep agents inside their analytical lens. **Enriched** profiles (Appendix C.22) add stricter output structure: numbered claims (`C1`, ...) with typed `reasoning_type`, cited evidence IDs, assumptions, falsifiers, and confidence, where critiques must reference claims by label and each position must cite the claim IDs justifying it (Appendix C.18).

Scaffolding tiers 2–3 require each proposal to make at least one causal or risk-assessment claim, state what evidence would change the agent's belief, and name potential confounders, with explicit warnings against reverse causation, confounding, selection bias, post-hoc narrative construction, and survivorship bias. We flag one design property as a limitation on interpretation: the same causal hierarchy is encoded in both the generation prompts and the CRIT *Causal Alignment* rubric (Appendix C.19), so agents are partly scored against the scaffolding they were given. This makes CRIT gains from enriched prompting easier to obtain and is one reason we treat rubric improvement as insufficient evidence of better decisions.

Experiments below are organized by research question (RQ1–RQ3), with sub-experiments numbered within each (e.g. RQ1-E1).

## 3.2 RQ1: Do Reasoning Metrics Predict Financial Performance?

### 3.2.1 RQ1-E1: CRIT and Financial Performance

Across **210 runs** spanning three intervention ablations (A7 and A8, JSD-triggered; A10, CRIT-triggered; Appendix C.1), CRIT reasoning quality ($\bar{\rho}$) shows **no meaningful relationship with financial performance**. Pooled correlations are near zero, paired changes in reasoning do not predict changes in performance, and regressions controlling for design factors find no predictive signal from $\bar{\rho}$. These 210 runs reuse scenarios across conditions, so the observations are not independent and the $p$-values are not scenario-clustered; we read the correlations descriptively, as effect sizes near zero rather than as a significance test. Subgroup analysis in Appendix C.15 reveals two localized signals: value-agent reasoning negatively correlates with Sharpe, while risk-agent reasoning weakly correlates with returns, though the latter is not statistically significant.

Table 1 shows that across all experiments the debate system **underperforms a naïve equal-ticker-weight baseline**. Debate portfolios beat $1/N$ only **33% of the time** (binomial $p < 0.001$) and exhibit consistently lower Sharpe ratios. Despite the weak relationship between reasoning metrics and financial performance, Appendix C.16 shows reasoning quality itself is highly sensitive to prompt structure. **Structured reasoning prompts substantially improve CRIT scores**. Enriched prompts increase mean reasoning quality from **0.72 to 0.84** ($+17.7\%$, Cohen's $d \approx 2.0$). These gains are driven primarily by improvements in evidential grounding and causal reasoning (Appendix C.17).

| Analysis | r | p | n |
|---|---|---|---|
| $\bar{\rho}$ vs Sharpe | $+0.07$ | 0.29 | 210 |
| $\bar{\rho}$ vs Return | $+0.03$ | 0.70 | 210 |
| $\Delta\rho$ vs $\Delta$Sharpe (paired) | $-0.02$ | 0.87 | 105 |
| $\Delta\rho$ vs $\Delta$Return (paired) | $+0.01$ | 0.91 | 105 |

**Table 1:** Pearson correlation between reasoning quality and financial outcomes.

These gains are **front-loaded rather than iterative**: enriched prompts score highest in Round 1, then $\bar{\rho}$ declines across later rounds in every condition. Scaffolding improves initial reasoning quality; debate does not improve it further, and neither translates into better risk-adjusted returns.

These results answer RQ1 in the negative for this setting: CRIT reasoning scores are not predictive of financial performance on our scenario set. We are, however, still interested in whether debate itself improves reasoning or financial outcomes.

### 3.3 RQ2: Does Debate Improve Reasoning and Portfolio Performance?

#### 3.3.1 RQ2-E1: Baseline Prompting and Debate Experiments

We evaluate whether structured multi-agent debate improves reasoning quality and portfolio performance relative to single-agent decision-making. These experiments modify the debate configuration itself rather than introducing runtime control signals, isolating the effect of debate as a reasoning mechanism.

| Metric | (A) Single Agent Basic | (B) 3 Roles Mean, Basic | (C) 3 Roles Mean, Causal | (D) 3 Roles Mean, Enriched |
|---|---|---|---|---|
| Excess Return % (v SPY) | $0.77\% \pm 1.03\%$ | $0.65\% \pm 0.93\%^{C}$ | $1.12\% \pm 0.93\%^{BD}$ | $0.57\% \pm 0.91\%^{C}$ |
| Sharpe | $1.00 \pm 0.30$ | $0.99 \pm 0.30^{C}$ | $1.07 \pm 0.30^{B}$ | $1.04 \pm 0.30$ |
| Volatility | $0.22 \pm 0.01^{\mathbf{CD}}$ | $0.21 \pm 0.01^{\mathbf{D}}$ | $0.21 \pm 0.01^{\mathbf{AD}}$ | $0.18 \pm 0.01^{\mathbf{ABC}}$ |
| Max Drawdown % | $9.84\% \pm 0.97\%^{\mathbf{CD}}$ | $9.60\% \pm 0.93\%^{\mathbf{CD}}$ | $9.33\% \pm 0.90\%^{AB\mathbf{D}}$ | $8.29\% \pm 0.77\%^{\mathbf{ABC}}$ |
| Sortino | $1.76 \pm 0.46$ | $1.75 \pm 0.47^{C}$ | $1.87 \pm 0.47^{B}$ | $1.80 \pm 0.47$ |
| Num Positions | $10.77 \pm 1.47^{B}$ | $11.37 \pm 1.55^{ACD}$ | $10.83 \pm 1.41^{BD}$ | $10.17 \pm 1.21^{BC}$ |
| Uninvested Cash | $8.26\% \pm 1.09\%^{\mathbf{D}}$ | $8.13\% \pm 0.56\%^{\mathbf{D}}$ | $7.51\% \pm 0.57\%^{\mathbf{D}}$ | $16.99\% \pm 1.06\%^{\mathbf{ABC}}$ |
| Total Return % | $4.29\% \pm 1.62\%$ | $4.16\% \pm 1.52\%^{C}$ | $4.63\% \pm 1.54\%^{BD}$ | $4.08\% \pm 1.36\%^{C}$ |

**Table 2:** Mean over 35 quarter+tickers scenarios $\pm$ SEM. Superscripts indicate stat. sig. in paired t-test with the specified column(s). Unbolded: $p < 0.15$, **Bolded**: $p < 0.01$).

**Prompt baselines.** All comparisons below are paired across the same 35 scenarios, with 95% CIs from a bootstrap resampled at the scenario level (the independent sampling unit). Replacing a single 'basic'-prompt agent with the mean vote of minimal Value, Risk, and Technical agents increased diversification and trended volatility and drawdown down, but return and Sharpe down as well; none of these differences is distinguishable from zero (return $-0.12$pp, CI $[-0.73, +0.48]$, $p$=0.69; columns A–B of Table 2). **Adding causal scaffolding to the same three-role ensemble did raise return and Sharpe** relative to the basic ensemble (return $+0.47$pp, CI $[+0.09, +0.84]$, $p$=0.024, $d$=0.40; Sharpe $+0.08$, CI $[+0.01, +0.16]$, $p$=0.043, $d$=0.36; column C). We note this gain does *not* survive comparison against the single-agent baseline (return $+0.34$pp, CI $[-0.38, +1.06]$, $p$=0.37), so causal scaffolding is best read as recovering ground lost by ensembling rather than as beating a single agent. Enriched prompts (column D) produced the only large effects in this table: volatility $-0.04$ (CI $[-0.05, -0.02]$, $p$<0.001, $d$=$-1.02$) and max drawdown $-1.54$pp (CI $[-2.11, -1.02]$, $p$<0.001, $d$=$-0.92$) versus the single agent, partly attributable to a rise in uninvested cash from $\sim$8% to 17%; return trended lower while Sharpe was largely retained. Structured prompting therefore reliably reduces *risk*; its effect on return is small and, against the strongest baseline, not statistically distinguishable from zero.

**Run-to-run noise floor.** Each run is a single draw at temperature 0.3. Re-running the column-A configuration six times on each of five scenarios (30 runs, everything else fixed) gives a pooled within-scenario SD of 1.30pp in return, 4–16$\times$ any between-condition difference in Table 2. Over 35 paired scenarios that implies a standard error of 0.31pp, plotted against the observed effects in Figure 1: every return difference falls inside it. The measured return differences are therefore small relative to run-to-run variability under these configurations, giving no evidence of a practically large return effect; the drawdown effect is the only one clearly above this floor. One configuration only; debate configurations make more calls and may be noisier (Appendix C.3).

**Causal debate baselines.** Continuing with the Causal prompt, adding a single round of critique, revise, and final judgement led to significant improvements in volatility and drawdown partially attributed to additional cash, though return and Sharpe trended lower (Appendix C.2 column B).

**Enriched debate baselines (RQ2-E2).** Debate underperformed the simple average under causal prompts, so we also examined enriched prompts and longer debates on three challenge scenarios. At $N$=2–3 the intervals are wide and overlapping, so we report these in Appendix C.4 as hypothesis-generating rather than alongside the 35-scenario results.

No configuration consistently outperforms SPY there either. The 3-agent enriched debate shows the lowest drawdown (5.33% vs. 9.73%) and highest Sharpe (0.76 vs. 0.35), but at $N$=3 with SEMs of

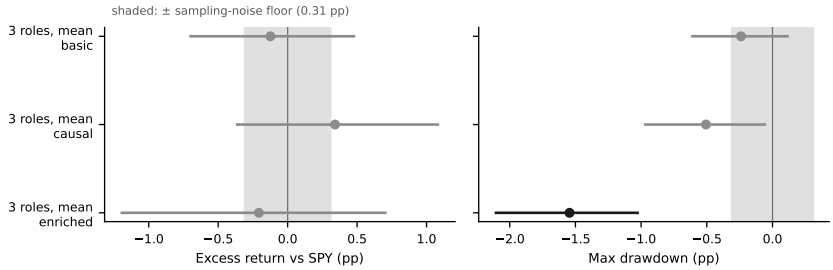



**Figure 1:** Paired differences against the single-agent baseline (Table 2, column A) over the same 35 scenarios, with 95% scenario-level bootstrap CIs. Shading is the sampling-noise floor from Appendix C.3. Every return difference lies inside it; only the enriched-prompt drawdown reduction clears it. Dark intervals exclude the band.

$\pm 0.71$ and $\pm 0.48$ these gaps are not separable from zero; they point the same way as the 35-scenario result and we treat them as consistent with, not independent evidence for, it. The 4-agent 5-round debate returns $-0.27\%$ while holding $\bar{\rho} \approx 0.83$–$0.85$ (Appendix C.6), the clearest single instance of high rubric scores alongside poor outcomes.

**Cross-provider robustness (RQ2-E3).** To test whether these findings depend on a particular model family, we repeated debates with mixed GPT/Gemini and GPT/Anthropic role assignments and with phase-level model routing. Returns were comparable across assignments whenever runs completed, and the dominant failure mode (agents acknowledging critiques without materially revising) recurred under every provider combination, so we treat the proxy–outcome findings as not specific to one model family. These runs are small ($N$=2–17) and one configuration failed in 3 of 6 scenarios when Gemini-2.5-flash served as judge and hallucinated a ticker outside the scenario universe; we therefore report them as supporting rather than confirmatory evidence, with full per-scenario results in Appendix C.8–C.11.

### 3.4 RQ3: Can Interventions Improve Reasoning and Portfolio Performance?

In this section we investigate whether targeted reasoning interventions can repair these failures during the debate process. Unlike the experiments in Section 3.3 (RQ2), which modify the debate configuration, these interventions operate dynamically by detecting reasoning failures and triggering corrective actions such as response retries or debate regulation.

#### 3.4.1 RQ3-E1: Repairing Consensus Convergence via RCA

Introducing an RCA-style consistency check in the revise phase of our causal-prompt debates reduced volatility and drawdown relative to the unregulated mean vote, while leaving return and Sharpe unchanged (column C of Table 3; paired over the same 35 scenarios). The risk reductions are the clearest effects in that table: volatility $-0.02$ (CI $[-0.02, -0.01]$, $p$<0.001, $d$=$-0.88$) and max drawdown $-0.69$pp (CI $[-1.09, -0.30]$, $p$=0.002, $d$=$-0.56$), whereas return ($-0.32$pp, CI $[-0.84, +0.20]$, $p$=0.25) and Sharpe ($-0.05$, CI $[-0.14, +0.04]$, $p$=0.31) are indistinguishable from zero. **Debate regulated by an RCA-style check therefore reduced risk without improving returns** on this scenario set: an investor optimizing raw return would still prefer the simple average of causally-scaffolded agents.

#### 3.4.2 RQ3-E2: Repairing Consensus Convergence via JSD

We also tried preventing consensus convergence (Section 3.5) and preserving disagreement by introducing a JSD collapse trigger at the post-revision checkpoint (between *revise* and *CRIT*). If the collapse ratio $JS_{\text{rev}}/JS_{\text{prop}}$ falls below threshold, role-specific nudge prompts are injected and the *revise* phase is re-executed before the updated portfolios proceed to CRIT and judging.

Appendix C.12 and Appendix C.13 report the full financial metrics. This is the only intervention producing significant risk-adjusted gains: Sharpe $+0.14$ ($t$=2.30, $p$=0.028, $d$=0.39) and Sortino $+0.25$ ($p$=0.026), with raw return $+0.83$pp not crossing the conventional threshold ($p$=0.052). Max drawdown is unchanged, so the gain comes from return generation rather than risk attenuation. Crucially, the successful configuration paired the trigger with a hard cap on revision magnitude (max 10% portfolio change); softer nudges had no measurable effect, so the trigger and the constraint are confounded and we cannot attribute the gain to disagreement preservation alone. **A debate**

**intervention can improve risk-adjusted performance under this two-agent configuration and scenario set**, though its active ingredient is not isolated.

Adding a third risk-specialist agent under the same trigger (A8) collapses the effect (Sharpe diff $+0.04$, $p = 0.53$; Appendix C.12), which we examine in Section 3.5.

### 3.4.3 RQ3-E3: Repairing Causal Rung Collapse

This experiment tests whether explicitly repairing causal reasoning failures improves portfolio performance. Instead of monitoring disagreement, the intervention targets the *causal alignment* pillar of the CRIT reasoning audit. After the CRIT phase, agents whose causal reasoning score $\rho$ falls below a predefined threshold receive targeted prompt nudges and re-run both *revise* and *CRIT* phases. This intervention therefore attempts to regulate reasoning quality directly rather than debate dynamics. Although intervention successfully improves causal reasoning scores, this, again, does not translate into financial gains. In the two-agent configuration (A10), Sharpe decreases slightly ($-0.09$, $p = 0.22$) and max drawdown increases (+0.53pp, $p = 0.062$). We discuss the likely reason, the technical agent's non-causal reasoning paradigm, in Section 4.

## 3.5 Failure Modes of Debate

**Consensus convergence.** The most consistent failure mode is *consensus convergence*: agents abandon their distinct investment theses during the critique-revise cycle. We use the term operationally, for a revision in which (i) an agent's allocation moves substantially toward the group consensus, (ii) the revision introduces no new evidence ID not already cited, and (iii) no identifiable factual or reasoning error in the agent's prior position was corrected. Where all three hold and the outcome does not improve, we describe the pattern as *consistent with sycophancy*; we do not claim to have isolated a social-deference mechanism, since legitimate error correction can also produce convergence and our design does not separate the two. In the 4-agent 5-round debate, cash allocations converge from 10–50% at initial proposal to 55–61% by round 5, and JSD between agent portfolios drops sharply after the first revision. The largest JSD drop occurs between the initial proposal and the first revision: by the time PID adjusts $\beta$ in subsequent rounds, the damage is done; thus PID prompt-based interventions (adjusting agreeableness $\beta$ based on $\bar{\rho}$ and JS divergence) have minimal practical effect. Even with aggressive anti-convergence prompts, agents converge to similar high-cash portfolios. This suggests the pattern is embedded in the critique-revise interaction itself and is not adequately controlled by prompt modulation alone. Shorter debates, and interventions that forcibly slow convergence, partially mitigate this by giving agents less opportunity to converge, preserving more portfolio diversity while retaining some of the risk-lowering benefits of structured critique.

**Causal alignment is the weakest pillar.** Across all experiments, $P_4$ (Causal Alignment) is consistently the lowest-scoring CRIT pillar ($0.65$–$0.81$), with `causal_overreach` flagged in most rounds. The Technical agent is the most frequent offender, making L2 (interventional) or L3 (counterfactual) claims supported only by L1 (associational) evidence on Pearl's causal ladder [Pearl, 2009]. This aligns with the RAudit framework's prediction that causal reasoning is the hardest dimension to sustain under debate pressure.

**Other failure modes.** **Ticker Hallucination.** When Gemini-2.5-flash serves as judge, it hallucinated a ticker (JNJ) not present in the scenario universe in 3 of 6 runs, causing those runs to produce invalid allocations. **Silent Prompt and Template Bugs.** Missing template variables and import errors caused incomplete prompts and hidden quality degradation. These were discovered through the dashboard file explorer, not through runtime errors: the system continued operating with degraded inputs rather than failing.

## 3.6 Qualitative Failure Case Studies

To ground the quantitative null results in mechanistic terms, we examine three concrete failure patterns identified through trace analysis of ablation logs.

**High CRIT, negative returns: convergent portfolio collapse.** The 4-agent, 5-round enriched debate achieves the highest reasoning quality scores across all configurations ($\bar{\rho} = 0.83$–$0.85$) yet produces negative average returns ($-0.27\%$, Sharpe $-0.05$). Trace inspection reveals the mechanism: cash allocations converge from 10–50% at initial proposal to 55–61% by round 5, and JSD between agent portfolios drops sharply after the first revision. Crucially, **the CRIT scorer evaluates the revised arguments after consensus convergence has already occurred**; a well-structured argument

for a high-cash allocation can receive a high CRIT score even when that allocation reflects social capitulation rather than genuine inference. The metric certifies argument quality at a point in time but cannot distinguish independently-derived positions from positions copied from a dominant agent. This explains why $\bar{\rho}$ and Sharpe are uncorrelated across the 210-run dataset: CRIT measures the quality of the *output* reasoning but not whether that reasoning drove an *independent* allocation decision.

**Reasoning–allocation disconnect across providers.** In the Claude-3.5-Sonnet (macro) + GPT-5-mini (risk, judge) configuration across 17 scenarios, trace analysis identifies **reasoning–allocation disconnect as the dominant failure in 10 of 17 runs**: agents produced internally coherent, well-scored arguments ($\rho_i$ high) but the judge's final portfolio allocation did not follow from those stated positions. In 7 runs, critique-ignoring compounded the disconnect—agents acknowledged critique points yet subsequent revisions were structurally unchanged. The macro agent was most frequently the weakest link, exhibiting causal overreach ($P_4$ failures) that destabilized the downstream allocation decision. The `position_rationale` linkage, which requires positions to cite supporting claim IDs, provides a structural safeguard but does not prevent a judge from discarding the debate's reasoning trail in favor of a simpler heuristic. Detecting this disconnect requires trace inspection; it is invisible to aggregate CRIT scores.

**When intervention succeeds: targeted signal preservation.** The only statistically significant performance gain in the study (A7 JSD intervention: Sharpe $+0.14$, $p = 0.028$, $d = 0.39$) illuminates the conditions under which debate repair succeeds. The two-agent baseline (macro + technical) exhibits a specific failure mode: the technical agent systematically converges toward the macro agent's position during revision, driving JSD collapse. It pairs a hard allocation constraint (max 10% change per revision) with targeting of the specific pair driving convergence. Adding a third agent (A8) collapses the effect ($+0.04$ Sharpe, $p = 0.53$): convergence spreads and the trigger loses its target. **Debate interventions must be designed around the precise failure mechanism**: the same intervention is the sole positive result in one configuration and null in another.

# 4 Conclusions

Across all three research questions we find reasoning behavior to be highly controllable—through prompting, debate structure, and interventions—while improvements in reasoning metrics rarely translate into improved financial performance.

**Why reasoning metrics do not predict returns (RQ1). CRIT and agreement measure coherence, not comprehensive correctness.** CRIT evaluates whether an argument is well-structured (logically valid, evidentially grounded, causally explicit) rather than whether its conclusion is *true*. Enriched prompts raise $\bar{\rho}$ from 0.72 to 0.84 without improving returns: agents learn to *write* better arguments, not to *make* better predictions. Rubrics of this kind work best in deterministic domains; in stochastic, partially-observable ones a bulletproof argument can be a bad bet. Holding cash to earn the risk-free rate is agreeable and near-unfalsifiable, and scores well, but does not maximize expected risk-adjusted return. Process quality is necessary but insufficient here.

**Public-information reasoning faces a structural ceiling.** Our agents reason exclusively over public information (SEC filings, news, and prices). Under even weak-form market efficiency, this information is largely incorporated into prices shortly after it becomes available. In this setting, improved reasoning about the same information cannot easily generate alpha. The empirical result that debate portfolios beat a naive $1/N$ allocation only 33% of the time is consistent with the well-known difficulty of outperforming equal-weight benchmarks [DeMiguel et al., 2009]. The negative result for RQ1 may therefore reflect a fundamental ceiling on the predictive value of public-information reasoning.

**Our simulation captures selection, not timing.** Quarterly rebalancing without transaction costs removes the dimensions where reasoning might matter most: *when* and *how* to trade. Portfolios with similar holdings perform similarly over a quarter regardless of reasoning quality. A different task design could therefore yield a different answer to RQ1.

**Signal diversity as a candidate mechanism for debate value (RQ2).** Debate alters reasoning behavior and portfolio characteristics, but its benefits appear to arise through signal diversity rather than improved reasoning accuracy: across our experiments, mechanisms preserving disagreement improved outcomes while mechanisms enforcing deeper reasoning did not. This is what ensemble diversity theory predicts—averaged-ensemble error equals mean member error *minus* a diversity

term [Krogh and Vedelsby, 1994]—and our system is such an ensemble. The critique–revise cycle introduces correlation because agents condition on each other's positions, so consensus convergence collapses allocations together and spends the diversity term; Condorcet's jury theorem makes the same point, that voting helps only while voters stay independent. We therefore read multi-agent debate here as a mechanism for regulating signal diversity rather than for improving reasoning quality. We note this is an interpretation consistent with our results rather than a directly tested mechanism: we did not manipulate diversity independently of the intervention.

**Limits of causal evaluation in financial reasoning (RQ3).** Interventions can modify reasoning behavior without reliably improving performance. Causal Alignment ($P_4$) is consistently the weakest pillar, yet explicitly repairing it (A10) slightly *degrades* outcomes—challenging the assumption that deeper causal reasoning is universally beneficial here. Technical analysis rests on statistical patterns (momentum, mean reversion, volume) that are empirically useful [Jegadeesh and Titman, 1993] but fundamentally associative: momentum does not *cause* price increase. Forcing such signals into higher-rung causal language produces "causal overreach", so the technical agent scores low on $P_4$ not because it reasons badly but because its paradigm is non-causal, while macro reasoning is natively causal ("CPI surprise $\rightarrow$ Fed tightening $\rightarrow$ rate-sensitive underperformance"). A uniform causal criterion conflates two legitimate reasoning styles; evaluation frameworks should allow role-specific metrics: causal coherence for macro agents, statistical grounding for technical ones.

Across all three questions the dominant driver of performance is the preservation of independent informational signals across agents, not improvement in individual reasoning quality. For practitioners building verifiers, the implication is narrow but concrete: a rubric that scores an agent's argument in isolation cannot detect that the argument was adopted rather than derived, and so cannot substitute for an external outcome check when one is available.

# 5 Limitations and Future Work

## 5.1 Limitations

**Statistical power.** Many configurations ran at $N{=}2$–3 scenarios, giving wide intervals; those results are exploratory. Only the 35-scenario paired comparisons support the significance claims we make; a direct run-to-run probe (Appendix C.3) confirms single-scenario comparisons at these effect sizes are uninterpretable. Extended debate costs 7–13 minutes and \$5–\$8 per scenario. **Outcome noise.** Realized return is a noisy signal, not ground-truth reasoning correctness: a well-reasoned allocation can lose money and a poorly-reasoned one can profit. Our null result for RQ1 shows CRIT does not predict this outcome; it does not show CRIT measures nothing. **Possible pre-training leakage.** Scenarios use historical data that may overlap LLM training corpora, so models may recall which stocks performed well, rewarding agents that reach the right answer without valid justification. **Limited realism.** A small ticker universe over a handful of quarters, with no transaction costs, spreads, taxes, or liquidity constraints, and no dynamic tool use: agents read precomputed memos and cannot compute or fetch live data. **Rubric–prompt coupling.** The causal hierarchy appears in both the generation scaffolding and the CRIT rubric, so enriched agents are partly graded on the structure they were handed. **Stratified analysis not executed.** A role- and regime-stratified re-analysis is implemented and unit-tested but has not been run against the full corpus; we report no stratified results. **Uneven reproducibility.** Every cell of Tables 2 and 3 regenerates exactly from stored per-scenario artifacts, as do the bootstrap CIs above. Three gaps remain. The configuration files for two of six conditions (causal mean vote, RCA-regulated debate) no longer match any surviving config by hash, so those runs can be re-analyzed but not re-executed byte-identically. The in-loop RCA retry is absent from our version history: the release contains only a post-hoc consistency evaluator, never invoked from the debate loop. It did run (all 35 scenarios differ from the unregulated condition) but only as an uncommitted working copy. And the per-round CRIT logs behind the $n{=}210$ correlations, the pillar tables, and the $\bar{\rho}\,0.72 \rightarrow 0.84$ gain are not in the released artifact set, so those numbers rest on author records rather than a re-runnable pipeline. RQ1 is our central claim and carries this caveat.

## 5.2 Broader Impact

That reasoning scores and consensus can both be maximized while performance stagnates cautions against treating them as sufficient evidence of good decision-making in deployed financial agents. Similar convergence across independently deployed systems could amplify correlated positioning under stress rather than supplying the diversity such systems are assumed to provide, a risk inferred from simulation.

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# A Appendix - Future Work

**Precise Sycophancy Intervention.** Refine debate prompts to maintain diversity and RAudit/RCA interventions to trigger highly selectively, before portfolio diversity collapses. Refine retry mechanisms that force revision when JSD drops too quickly may address convergence before it becomes irreversible. Develop a version of CRIT more appropriate for the domain.
**Scale for Statistical Significance.** Run each configuration across 50+ scenarios to narrow confidence intervals and enable meaningful hypothesis tests on metrics. Expanding the ticker universe to 50-100+ tickers would test whether debate-based selection provides value closer to real portfolio construction.
**Temporal and Regime Coverage.** Run simulations across consecutive quarters with realistic spreads and tax implications so agents can observe consequences of past decisions, incorporate regret signals, and learn. Systematically evaluate across bull, bear, and high-volatility regimes to assess regime-dependent behavior beyond our current sample.
**Distillation.** If our strongest configurations–3-agent causal mean, and 3-agent intervened debate–risk-adjusted advantage holds at scale, distill their most effective constraints- structured claims and point-by-point critique- into single-agent prompts to capture multi-agent benefits at lower latency and cost.

# B Appendix - System Diagram

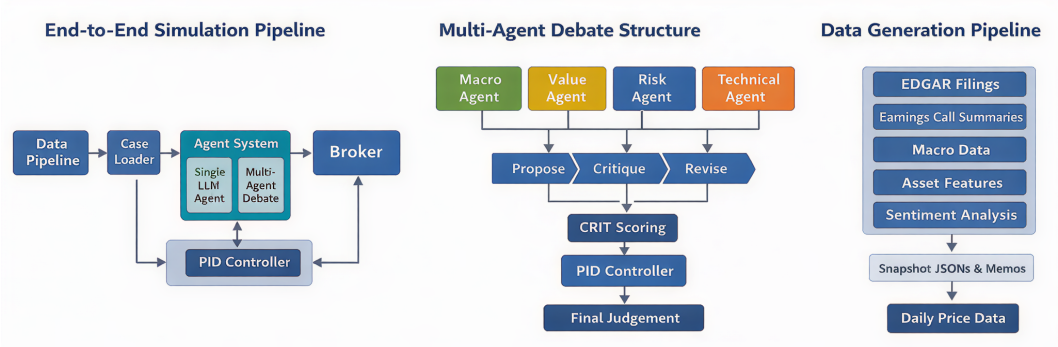



**Figure 2:** Our simulation pipeline builds data cases from input financial sources, orchestrates multi-agent debate in a market environment, then scores the decision-making through financial performance and reasoning quality (left). LLM agents with clear roles participate in Propose-Critique-Revise debate loops. The revision tone is adjusted by PID control (center). Input sources are summarized and formatted into quarterly snapshot memos Appendix C.23.

# C Appendix - Tables

## C.1 CRIT and Financial Performance

```
======================================================================
CRIT FINANCIAL CORRELATION ANALYSIS
======================================================================
Date: 2026-03-12 12:35
Total runs loaded: 210
  Ablation 7: 70 runs (baseline=35, treatment=35)
  Ablation 8: 70 runs (baseline=35, treatment=35)
  Ablation 10: 70 runs (baseline=35, treatment=35)

Descriptive statistics:
          rho_bar: mean=0.8371  std=0.0227  min=0.7275  max=0.8825
           sharpe: mean=0.5425  std=1.6479  min=-2.5705  max=4.8684
     total_return: mean=2.0813  std=8.5611  min=-19.0297  max=24.9141

======================================================================
1. POOLED CORRELATIONS (all runs)
======================================================================
```

```
  rho_bar vs Sharpe  (n=210)
    Pearson  r=+0.0735  p=0.2887
    Spearman r=+0.0481  p=0.4879

  rho_bar vs TotalReturn  (n=210)
    Pearson  r=+0.0265  p=0.7030
    Spearman r=+0.0139  p=0.8415


======================================================================
2. WITHIN ABLATION CORRELATIONS
======================================================================

--- Ablation 7 ---
  rho_bar vs Sharpe  (n=70)
    Pearson  r=+0.0007  p=0.9954
    Spearman r=-0.0444  p=0.7152

  rho_bar vs TotalReturn  (n=70)
    Pearson  r=-0.0539  p=0.6579
    Spearman r=-0.0749  p=0.5376


--- Ablation 8 ---
  rho_bar vs Sharpe  (n=70)
    Pearson  r=+0.1568  p=0.1949
    Spearman r=+0.0632  p=0.6030

  rho_bar vs TotalReturn  (n=70)
    Pearson  r=+0.1494  p=0.2172
    Spearman r=+0.0559  p=0.6457


--- Ablation 10 ---
  rho_bar vs Sharpe  (n=70)
    Pearson  r=+0.0865  p=0.4765
    Spearman r=+0.1104  p=0.3631

  rho_bar vs TotalReturn  (n=70)
    Pearson  r=+0.0287  p=0.8134
    Spearman r=+0.0413  p=0.7344


======================================================================
3. WITHIN CONDITION CORRELATIONS
======================================================================

--- Condition: baseline ---
  (n=105 runs)
  rho_bar vs Sharpe  (n=105)
    Pearson  r=+0.0042  p=0.9662
    Spearman r=+0.0042  p=0.9660

  rho_bar vs TotalReturn  (n=105)
    Pearson  r=-0.0285  p=0.7730
    Spearman r=-0.0260  p=0.7922


--- Condition: treatment ---
  (n=105 runs)
  rho_bar vs Sharpe  (n=105)
    Pearson  r=+0.1353  p=0.1689
    Spearman r=+0.0931  p=0.3448

  rho_bar vs TotalReturn  (n=105)
```

```
    Pearson  r=+0.0728  p=0.4607
    Spearman r=+0.0508  p=0.6069


--- Within ablation x condition ---
  Abl 7, baseline (n=35):
  rho_bar vs Sharpe  (n=35)
    Pearson  r=-0.1038  p=0.5529
    Spearman r=-0.0659  p=0.7069

  rho_bar vs TotalReturn  (n=35)
    Pearson  r=-0.0956  p=0.5849
    Spearman r=-0.0922  p=0.5982

  Abl 7, treatment (n=35):
  rho_bar vs Sharpe  (n=35)
    Pearson  r=+0.0738  p=0.6736
    Spearman r=+0.0070  p=0.9681

  rho_bar vs TotalReturn  (n=35)
    Pearson  r=-0.0275  p=0.8755
    Spearman r=-0.0575  p=0.7430

  Abl 8, baseline (n=35):
  rho_bar vs Sharpe  (n=35)
    Pearson  r=+0.0533  p=0.7609
    Spearman r=+0.0168  p=0.9236

  rho_bar vs TotalReturn  (n=35)
    Pearson  r=+0.0734  p=0.6752
    Spearman r=-0.0118  p=0.9465

  Abl 8, treatment (n=35):
  rho_bar vs Sharpe  (n=35)
    Pearson  r=+0.2755  p=0.1092
    Spearman r=+0.1966  p=0.2578

  rho_bar vs TotalReturn  (n=35)
    Pearson  r=+0.2384  p=0.1679
    Spearman r=+0.1964  p=0.2581

  Abl 10, baseline (n=35):
  rho_bar vs Sharpe  (n=35)
    Pearson  r=+0.0927  p=0.5965
    Spearman r=+0.1735  p=0.3188

  rho_bar vs TotalReturn  (n=35)
    Pearson  r=-0.0092  p=0.9581
    Spearman r=+0.0726  p=0.6785

  Abl 10, treatment (n=35):
  rho_bar vs Sharpe  (n=35)
    Pearson  r=+0.1059  p=0.5449
    Spearman r=+0.0540  p=0.7582

  rho_bar vs TotalReturn  (n=35)
    Pearson  r=+0.0858  p=0.6243
    Spearman r=+0.0142  p=0.9357


======================================================================
4. WITHIN SCENARIO PAIRED DIFFERENCES (delta analysis)
======================================================================
For each scenario appearing in both baseline and treatment within an ablation,
compute delta_rho = treatment.rho_bar - baseline.rho_bar
```

```
        delta_sharpe = treatment.sharpe - baseline.sharpe
        delta_return = treatment.total_return - baseline.total_return
Then correlate the deltas.

Paired scenarios found: 105
  Ablation 7: 35 pairs
  Ablation 8: 35 pairs
  Ablation 10: 35 pairs


--- Pooled deltas ---
  delta_rho:    mean=-0.0003  std=0.0329
  delta_sharpe: mean=+0.0327  std=0.4072
  delta_return: mean=+0.2622  std=2.0966

  delta_rho vs delta_Sharpe  (n=105)
    Pearson  r=-0.0157  p=0.8738
    Spearman r=+0.0066  p=0.9465

  delta_rho vs delta_Return  (n=105)
    Pearson  r=+0.0114  p=0.9084
    Spearman r=+0.0331  p=0.7373


--- Deltas within ablation 7 ---
  n=35 pairs
  delta_rho:    mean=+0.0013  std=0.0382
  delta_sharpe: mean=+0.1432  std=0.3684
  delta_return: mean=+0.8285  std=2.4367

  delta_rho vs delta_Sharpe  (n=35)
    Pearson  r=-0.1270  p=0.4673
    Spearman r=-0.1219  p=0.4856

  delta_rho vs delta_Return  (n=35)
    Pearson  r=-0.0690  p=0.6935
    Spearman r=-0.0521  p=0.7663


--- Deltas within ablation 8 ---
  n=35 pairs
  delta_rho:    mean=-0.0139  std=0.0234
  delta_sharpe: mean=+0.0439  std=0.4085
  delta_return: mean=+0.1557  std=1.8887

  delta_rho vs delta_Sharpe  (n=35)
    Pearson  r=-0.0109  p=0.9505
    Spearman r=+0.0132  p=0.9402

  delta_rho vs delta_Return  (n=35)
    Pearson  r=+0.0720  p=0.6809
    Spearman r=+0.0565  p=0.7474


--- Deltas within ablation 10 ---
  n=35 pairs
  delta_rho:    mean=+0.0116  std=0.0312
  delta_sharpe: mean=-0.0890  std=0.4203
  delta_return: mean=-0.1975  std=1.8389

  delta_rho vs delta_Sharpe  (n=35)
    Pearson  r=+0.2072  p=0.2324
    Spearman r=+0.2217  p=0.2005

  delta_rho vs delta_Return  (n=35)
```

```
    Pearson  r=+0.1545  p=0.3755
    Spearman r=+0.1266  p=0.4685


================================================================================
5. PER AGENT RHO CORRELATIONS
================================================================================
Agents found across dataset: macro, risk, technical


--- Agent: macro ---
  n=210 runs with rho_macro
  rho_macro vs Sharpe  (n=210)
    Pearson  r=+0.0377  p=0.5866
    Spearman r=+0.0212  p=0.7602

  rho_macro vs TotalReturn  (n=210)
    Pearson  r=+0.0158  p=0.8202
    Spearman r=+0.0017  p=0.9810


--- Agent: risk ---
  n=70 runs with rho_risk
  rho_risk vs Sharpe  (n=70)
    Pearson  r=+0.0083  p=0.9453
    Spearman r=+0.0027  p=0.9826

  rho_risk vs TotalReturn  (n=70)
    Pearson  r=+0.0405  p=0.7393
    Spearman r=+0.0084  p=0.9448


--- Agent: technical ---
  n=210 runs with rho_technical
  rho_technical vs Sharpe  (n=210)
    Pearson  r=+0.0820  p=0.2366
    Spearman r=+0.0706  p=0.3086

  rho_technical vs TotalReturn  (n=210)
    Pearson  r=+0.0274  p=0.6932
    Spearman r=+0.0464  p=0.5033


--- Agent rho comparison table (Pearson r with Sharpe) ---
  Agent                  r        p     n
  ---------------------------------------
  macro            +0.0377   0.5866   210
  risk             +0.0083   0.9453    70
  technical        +0.0820   0.2366   210


--- Agent rho comparison table (Pearson r with TotalReturn) ---
  Agent                  r        p     n
  ---------------------------------------
  macro            +0.0158   0.8202   210
  risk             +0.0405   0.7393    70
  technical        +0.0274   0.6932   210

================================================================================
6. OLS REGRESSION: Sharpe ~ rho_bar + ablation dummies
================================================================================
Controls for ablation level effects using dummy variables.
Reference ablation: the lowest numbered ablation.
```

```
--- Dependent variable: Sharpe (ref ablation=7) ---
  n=210, R2=0.0072, Adj R2=-0.0072
  F(3,206)=0.499, p=0.6832
  Variable                         Coef         SE        t        p
  ------------------------------------------------------------------
  (intercept)                   -3.8738     4.2549   -0.910   0.3637
  rho_bar                        5.3388     5.1030    1.046   0.2967
  abl_8_dummy                   -0.1515     0.2803   -0.540   0.5895
  abl_10_dummy                  -0.0064     0.2829   -0.023   0.9820


--- Dependent variable: TotalReturn (ref ablation=7) ---
  n=210, R2=0.0028, Adj R2=-0.0117
  F(3,206)=0.196, p=0.8993
  Variable                         Coef         SE        t        p
  ------------------------------------------------------------------
  (intercept)                   -5.4973    22.1532   -0.248   0.8043
  rho_bar                        9.2436    26.5692    0.348   0.7283
  abl_8_dummy                   -0.7028     1.4593   -0.482   0.6306
  abl_10_dummy                   0.2261     1.4732    0.153   0.8782


--- Extended model: Sharpe ~ rho_bar + ablation dummies + condition dummy ---
  n=210, R2=0.0073, Adj R2=-0.0120
  F(4,205)=0.378, p=0.8239
  Variable                         Coef         SE        t        p
  ------------------------------------------------------------------
  (intercept)                   -3.8961     4.2675   -0.913   0.3623
  rho_bar                        5.3447     5.1153    1.045   0.2973
  abl_8_dummy                   -0.1515     0.2810   -0.539   0.5904
  abl_10_dummy                  -0.0064     0.2836   -0.023   0.9819
  treatment_dummy                0.0345     0.2288    0.151   0.8801


--- Extended model: TotalReturn ~ rho_bar + ablation dummies + condition dummy ---
  n=210, R2=0.0031, Adj R2=-0.0164
  F(4,205)=0.158, p=0.9590
  Variable                         Coef         SE        t        p
  ------------------------------------------------------------------
  (intercept)                   -5.6681    22.2177   -0.255   0.7989
  rho_bar                        9.2893    26.6315    0.349   0.7276
  abl_8_dummy                   -0.7030     1.4627   -0.481   0.6313
  abl_10_dummy                   0.2257     1.4766    0.153   0.8787
  treatment_dummy                0.2654     1.1912    0.223   0.8239


======================================================================
7. SUMMARY AND INTERPRETATION
======================================================================
Total observations: 210
Pooled rho_bar Sharpe:      Pearson r=+0.0735 (p=0.2887), Spearman r=+0.0481 (p
    =0.4879)
Pooled rho_bar TotalReturn: Pearson r=+0.0265 (p=0.7030)

No statistically significant pooled correlation between CRIT scores and Sharpe.

Delta analysis (paired within scenario): r=-0.0157 (p=0.8738)
Changes in CRIT quality do NOT significantly predict changes in financial
    performance.

Note: Correlations marked with * are significant at p < 0.05.
All p values are two tailed.
```

## C.2 Causal Debate Baselines & RCA

| Metric | (A) 3 Roles Mean, Causal | (B) 3 Roles Debate, Causal | (C) 3 Roles Debate, Causal + RCA |
|---|---|---|---|
| Excess Return % (v SPY) | $1.12\% \pm 0.93\%$ | $0.95\% \pm 0.96\%$ | $0.80\% \pm 0.88\%$ |
| Sharpe | $1.07 \pm 0.30$ | $1.01 \pm 0.30$ | $1.03 \pm 0.30$ |
| Volatility | $0.21 \pm 0.01^{\mathbf{BC}}$ | $0.20 \pm 0.01^{\mathbf{A}C}$ | $0.19 \pm 0.01^{\mathbf{A}B}$ |
| Max Drawdown % | $9.33\% \pm 0.90\%^{B\mathbf{C}}$ | $8.88\% \pm 0.83\%^{AC}$ | $8.64\% \pm 0.80\%^{\mathbf{A}B}$ |
| Sortino | $1.87 \pm 0.47$ | $1.82 \pm 0.49$ | $1.80 \pm 0.47$ |
| Num Positions | $10.83 \pm 1.41$ | $11.17 \pm 1.49$ | $11.14 \pm 1.57$ |
| Uninvested Cash | $7.51\% \pm 0.57\%^{\mathbf{BC}}$ | $12.63\% \pm 0.78\%^{\mathbf{A}C}$ | $13.80\% \pm 1.00\%^{\mathbf{A}B}$ |
| Total Return % | $4.63\% \pm 1.54\%$ | $4.47\% \pm 1.49\%$ | $4.31\% \pm 1.41\%$ |

**Table 3:** Mean over 35 quarter+tickers scenarios $\pm$ SEM. Superscripts indicate stat. sig. in paired t-test with the specified column(s). Unbolded: $p < 0.15$, **Bolded**: $p < 0.01$).

## C.3 Run-to-Run Variance Probe

To quantify sampling noise at fixed configuration, we re-ran the `single_agent_slim` configuration (Table 2, column A) six times on each of five scenarios spanning distinct regimes, varying only the LLM sampling draw (temperature 0.3); configuration, scenario, and evaluation window were held constant. All 30 runs completed successfully.

| | Total return % | | | Annualized Sharpe | | |
|---|---|---|---|---|---|---|
| Scenario | mean | SD | range | mean | SD | range |
| 2022 Q1 | 2.29 | 1.25 | 3.17 | 0.566 | 0.285 | 0.739 |
| 2022 Q3 rand3 | 1.25 | 0.25 | 0.70 | 0.257 | 0.043 | 0.118 |
| 2022 Q4 tech | 1.37 | 0.59 | 1.43 | 0.229 | 0.077 | 0.194 |
| 2023 Q2 | 12.81 | 2.29 | 6.92 | 3.140 | 0.173 | 0.476 |
| 2025 Q1 tech | −16.47 | 1.11 | 3.25 | −1.982 | 0.091 | 0.280 |
| Pooled SD (one run) | | 1.30 | | | 0.159 | |
| Implied SEM, $n$=35 paired | | 0.31 | | | 0.038 | |

**Table 4:** Run-to-run variance at fixed configuration and scenario ($n$=6 repeats each). Within-scenario SD exceeds every between-condition difference in Table 2, so single-scenario comparisons between those configurations are uninformative. The implied paired standard error over 35 scenarios is the same order as those differences. Variance is heterogeneous across scenarios, so the pooled figure should be read as an approximation.

Reproduced by `analysis/seed_variance_runs.sh` and `analysis/seed_variance.py`; per-run artifacts are stored under `results/seed_variance/`.

## C.4 Enriched Debate Configurations (Small-$N$)

| **Configuration** | **N** | **Return %** | **Sharpe** | **Sortino** | **Max DD %** | **Excess vs SPY %** |
|---|---|---|---|---|---|---|
| Single Agent | 3 | $+1.59 \pm 0.99$ | $0.35 \pm 0.17$ | $0.52 \pm 0.25$ | $9.73 \pm 1.90$ | $-1.01 \pm 3.12$ |
| Multi-Agent Mean | 3 | $+2.43 \pm 1.23$ | $0.58 \pm 0.30$ | $0.88 \pm 0.46$ | $8.61 \pm 1.95$ | $-0.13 \pm 3.09$ |
| Enriched Debate (3-ag.) | 3 | $+2.46 \pm 1.74$ | $0.76 \pm 0.48$ | $1.16 \pm 0.75$ | $5.33 \pm 0.71$ | $+0.04 \pm 4.04$ |
| Enriched Debate (4-ag.) | 2 | $-0.27 \pm 0.29$ | $-0.05 \pm 0.06$ | $-0.07 \pm 0.09$ | $6.09 \pm 0.82$ | $+0.14 \pm 4.47$ |

**Table 5:** Financial performance across configurations (mean $\pm$ SEM). Excess vs SPY is portfolio return minus S&P 500 return over the same quarter. At $N$=2–3 these intervals are wide and overlapping, so the ordering is indicative only: the 3-agent enriched debate shows the highest Sharpe and Sortino and the lowest drawdown, while the 4-agent 5-round debate is the only configuration with negative mean return.

### C.5 Enriched Debate Scenario Breakdown

| Scenario | Return % | Sharpe | Sortino | Max DD % | SPY % | Excess % |
|---|---|---|---|---|---|---|
| 2022 Q4 Tech Sector | $-0.35$ | $-0.13$ | $-0.20$ | 5.96 | $+4.79$ | $-5.14$ |
| 2022 Q1 Inflation Shock | $+2.83$ | 0.89 | 1.32 | 6.11 | $-5.16$ | $+7.99$ |
| 2023 Q2 Higher-for-Longer | $+5.52$ | 1.52 | 2.37 | 3.91 | $+8.27$ | $-2.75$ |
| **Average** | $+2.67$ | 0.76 | 1.16 | 5.33 | $+2.63$ | $+0.04$ |

**Table 6:** Per-scenario results for 3-agent enriched debate (Config 3). Performance varies substantially across market regimes: the system outperforms SPY by $+7.99\%$ during the 2022 Q1 inflation shock but underperforms by $-5.14\%$ in the 2022 Q4 tech rally.

### C.6 Reasoning Quality

| Round | $\bar{\rho}$ | $P_1$ (LV) | $P_2$ (ES) | $P_3$ (AC) | $P_4$ (CA) |
|---|---|---|---|---|---|
| 1 | 0.843 | 0.86 | 0.86 | 0.87 | 0.78 |
| 2 | 0.845 | 0.87 | 0.85 | 0.86 | 0.81 |
| 3 | 0.815 | 0.84 | 0.83 | 0.80 | 0.75 |
| 4 | 0.832 | 0.77 | 0.86 | 0.88 | 0.81 |
| 5 | 0.839 | 0.82 | 0.86 | 0.86 | 0.79 |

**Table 7:** CRIT pillar scores across debate rounds (4-agent enriched debate, 2022 Q4 tech scenario). $P_1$=Logical Validity, $P_2$=Evidential Support, $P_3$=Alternative Consideration, $P_4$=Causal Alignment. The most common CRIT diagnostic is `causal_overreach`, particularly for the Technical agent.

### C.7 Cross-Provider Debate: Detailed Findings

This appendix preserves the full narrative for the cross-provider experiments summarized in the main text (RQ2-E3).

**GPT and Gemini.** With `gpt-5-mini` serving as Macro, Value, and Judge and `gemini-2.5-flash` as Risk, 6/6 scenarios completed successfully. With the assignment reversed, only 3/6 completed: Gemini as judge hallucinated ticker JNJ, which was not in the scenario universe, invalidating those allocations. Returns are comparable across assignments when runs complete, suggesting financial outcomes are largely provider-agnostic, while judge *reliability* is not (6/6 vs. 3/6 completion). Notably, all three configurations produced byte-identical portfolios on one scenario—same positions, same cash, same quarterly return—despite different providers and run times. This convergence to equal-weight allocation under market stress is consistent with the consensus-convergence dynamic described in Section 3.5 operating independently of the underlying model.

**Phase-level routing.** With `gpt-5-mini` handling propose and revise and `gemini-2.5-flash` handling critique and judge, the 2022 Q1 Inflation Shock scenario (10 tickers) returned $-16.22\%$ (Sharpe $-2.11$, Sortino $-2.63$, max drawdown $17.67\%$), an excess of $+0.13\%$ against SPY's $-16.35\%$ (Table 9).

**GPT and Anthropic.** With `claude-3.5-sonnet` as macro agent and `gpt-5-mini` as risk agent and judge across 17 scenarios (2021–2025, three debate rounds), mean return was $-1.11\%$ and mean excess return versus the S&P 500 was $+1.01\%$. Outcomes were highly dispersed ($+21.71\%$ to $-70.10\%$; Sharpe $-2.73$ to $4.38$; Appendix C.10). Trace analysis identified reasoning–allocation disconnect as dominant (10 of 17 runs) and critique-ignoring or stalled revision in 7 runs, with the macro agent most often flagged weakest for unsupported claims or causal overreach. Comparing role-only against hybrid phase-routing under the same three-round structure (Appendix C.11), excess returns versus SPY stayed positive ($+2.26\%$ to $+3.16\%$) despite negative absolute returns ($-2.37\%$ to $-1.42\%$); role-only with Claude as macro achieved the strongest excess return, while hybrid phase routing produced slightly lower Sharpe and higher final JS divergence, indicating less convergence. Given $N$=2–17 and the judge-hallucination failures, these comparisons are exploratory.

| Scenario | Exp | Return % | Sharpe | Sortino | Max DD % | Excess % | Status |
|---|---|---|---|---|---|---|---|
| 2021 Q3 Inflation | A | +12.40 | 3.05 | 4.65 | 4.61 | +2.64 | OK |
| | B | - | - | - | - | - | FAIL |
| 2022 Q1 Inf. Shock | A | −19.40 | −2.56 | −3.16 | 19.96 | −3.05 | OK |
| | B | −19.40 | −2.56 | −3.16 | 19.96 | −3.05 | OK |
| 2022 Q2 Recession | A | −1.97 | −0.35 | −0.50 | 12.69 | +3.96 | OK |
| | B | −1.77 | −0.32 | −0.45 | 12.97 | +4.16 | OK |
| 2022 Q4 Banking | A | +2.75 | 0.49 | 0.70 | 8.16 | −5.17 | OK |
| | B | - | - | - | - | - | FAIL |
| 2023 Q2 Higher4Lngr | A | −0.03 | −0.37 | −0.50 | 5.53 | +3.31 | OK |
| | B | - | - | - | - | - | FAIL |
| 2023 Q4 AI Boom | A | +16.91 | 4.66 | 9.24 | 2.04 | +5.90 | OK |
| | B | +14.42 | 4.43 | 8.40 | 1.91 | +3.41 | OK |
| **Mean (completed)** | A | +1.78 | 0.82 | 1.74 | 8.83 | +1.26 | 6/6 |
| | B | −2.25 | 0.52 | 1.60 | 11.61 | +1.51 | 3/6 |

**Table 8:** Per-scenario LLM comparison. Exp A uses GPT-5-mini for all roles; Exp B uses Gemini-2.5-flash for Macro/Value/Judge and GPT-5-mini for Risk. Exp B failures are caused by Gemini hallucinating ticker JNJ (not in scenario universe). Where both complete, returns are comparable-e.g., 2022 Q1 produces byte-identical portfolios.

### C.8 GPT and Gemini Experiments

### C.9 GPT and Gemini Phase LLM results

| Metric | Value |
|---|---|
| Scenario | 2022 Q1 Inflation Shock (10 tickers) |
| Return | −16.22% |
| Sharpe | −2.11 |
| Sortino | −2.63 |
| Max Drawdown | 17.67% |
| SPY Return | −16.35% |
| Excess vs SPY | +0.13% |

**Table 9:** Phase-LLM experiment results (GPT-5-mini for propose/revise, Gemini-2.5-flash critique/-judge). Tracked SPY closely during a severe downturn, with excess return of +0.13%.

### C.10 GPT and Anthropic Ablation Tests

| Metric | Mean | Best | Worst |
|---|---|---|---|
| Return (%) | -1.11 | 21.71 | -70.10 |
| Excess Return vs SPY (%) | 1.01 | 11.58 | -6.41 |
| Max Drawdown (%) | 7.87 | 1.68 | 16.39 |
| Sharpe Ratio | 1.06 | 4.38 | -2.73 |
| Trades | 8.8 | 21 | 1 |

**Table 10:** Ablation performance across 17 market scenarios.

## C.11 GPT and Anthropic Routing Experiments

| Experiment | Return (%) | Excess vs SPY (%) | Sharpe | Final JS |
|---|---|---|---|---|
| Exp 1 Role-only (Macro Anthropic, Risk GPT) | -1.42 | 3.16 | -0.54 | 0.7181 |
| Exp 2 Role-only (Macro GPT, Risk Anthropic) | -2.37 | 2.26 | -0.64 | 0.7312 |
| Exp 3 Hybrid phases | -1.70 | 2.84 | -0.53 | 0.7946 |

**Table 11:** Performance comparison across LLM routing experiments.

## C.12 Performance Metrics across Intervention Ablations

| Abl. | Config | N | Ret | Sharpe | Sortino | MaxDD | Excess vs SPY |
|---|---|---|---|---|---|---|---|
| A7 | Base | 35 | $+1.79 \pm 1.50$ | $+0.50 \pm 0.28$ | $+0.93 \pm 0.43$ | $+10.03 \pm 0.88$ | $-1.73 \pm 1.03$ |
| | +JS | 35 | $+2.62 \pm 1.61$ | $\mathbf{+0.64 \pm 0.29}$ | $\mathbf{+1.19 \pm 0.46}$ | $+10.21 \pm 0.95$ | $-0.90 \pm 1.08$ |
| A8 | Base | 35 | $+1.46 \pm 1.21$ | $+0.42 \pm 0.28$ | $+0.81 \pm 0.44$ | $+8.07 \pm 0.71$ | $-2.06 \pm 0.87$ |
| | +CRIT | 35 | $+1.61 \pm 1.26$ | $+0.46 \pm 0.28$ | $+0.86 \pm 0.43$ | $+8.27 \pm 0.78$ | $-1.90 \pm 0.94$ |
| A10 | Base | 35 | $+2.61 \pm 1.54$ | $+0.66 \pm 0.28$ | $+1.18 \pm 0.44$ | $+9.80 \pm 0.87$ | $-0.91 \pm 1.04$ |
| | +JS | 35 | $+2.41 \pm 1.59$ | $+0.57 \pm 0.27$ | $+1.04 \pm 0.42$ | $\mathbf{+10.33 \pm 0.93}$ | $-1.11 \pm 1.07$ |

**Table 12:** Performance metrics across intervention ablations. Three ablations evaluate reasoning interventions in the multi-agent debate system. **A7** introduces a Jensen–Shannon (JS) collapse detector in a two-agent setting (*macro*, *technical*), triggering when revision-stage disagreement falls below a threshold ($JS_{\text{rev}}/JS_{\text{prop}} < 0.8$) and nudging agents to maintain independent reasoning. **A8** applies the same JS intervention with an added *risk* agent (three agents total); the intervention fired frequently (97% of debates) but produced no measurable effect. **A10** replaces the JS trigger with a CRIT-based reasoning intervention that fires when causal reasoning quality ($\rho_4 < 0.8$).

## C.13 T-tests Results for Ablations

| Abl. | Metric | Diff | 95% CI | t | p | d |
|---|---|---|---|---|---|---|
| A7 | Sharpe | $+0.143$ | $[+0.017, +0.270]$ | $+2.300$ | **0.0277** | $+0.389$ |
| A7 | Sortino | $+0.254$ | $[+0.032, +0.476]$ | $+2.328$ | **0.0260** | $+0.393$ |
| A7 | Excess vs SPY % | $+0.828$ | $[-0.009, +1.666]$ | $+2.011$ | 0.0523 | $+0.340$ |
| A10 | Max DD % | $+0.529$ | $[-0.028, +1.086]$ | $+1.930$ | **0.0620** | $+0.326$ |

**Table 13:** Paired t-test results for ablations with statistically significant or marginal effects ($p < 0.10$). Green p-values indicate improvement relative to the baseline configuration, red indicates undesired outcomes.

## C.14 Prompt Enrichment Tiers

| Tier | n | $\bar{\rho}$ | Sharpe | Return (%) |
|---|---|---|---|---|
| Standard | 56 | 0.833 | -0.918 | -3.56 |
| Light | 62 | 0.825 | -0.940 | -3.58 |
| Intense Light | 67 | 0.819 | -0.992 | -3.66 |
| Intense | 60 | 0.834 | -0.967 | -3.75 |

**Table 14:** Prompt enrichment tiers (Standard $\rightarrow$ Intense) show no improvement in reasoning or financial performance across 245 runs of Ablation 1.

### C.15 Pearson correlation between Agent Reasoning Quality and Performance

| Subgroup | Metric | r | p | n |
|---|---|---|---|---|
| Value agent $\rho$ vs Sharpe | Pearson | $-0.29$ | 0.001 | 126 |
| Risk agent $\rho$ vs Return | Pearson | $+0.22$ | 0.006 | 152 |

**Table 15:** Selected subgroup correlations between agent reasoning quality and performance.

### C.16 Debate Portfolio Performance

| Ablation | n | Debate Return | EW Return | Diff | Beats EW |
|---|---|---|---|---|---|
| Abl 1 | 245 | $-3.64\%$ | $-3.28\%$ | $-0.36\%$ | 31% |
| Abl 7 | 70 | $+2.20\%$ | $+4.47\%$ | $-2.26\%$ | 30% |
| Abl 8 | 70 | $+1.54\%$ | $+4.47\%$ | $-2.93\%$ | 34% |
| Abl 10 | 70 | $+2.51\%$ | $+4.47\%$ | $-1.96\%$ | 40% |
| **Pooled** | 455 | — | — | $\mathbf{-1.29\%}$ | **33%** |

**Table 16:** Debate portfolio performance compared to an equal-weight baseline.

### C.17 Improvements in CRIT

| Pillar | Baseline | Enriched | $\Delta$ | Cohen's $d$ |
|---|---|---|---|---|
| Evidential Support | 0.51 | 0.85 | $+0.34$ | 2.92 |
| Causal Alignment | 0.70 | 0.87 | $+0.17$ | 2.21 |
| Alternative Consideration | 0.79 | 0.89 | $+0.10$ | 1.09 |
| Logical Validity | 0.87 | 0.82 | $-0.05$ | $-0.44$ |

**Table 17:** Improvements in CRIT reasoning pillars under structured prompts.

### C.18 Structured Claim–Evidence Schema

**Structured Claim-Evidence Schema** As mentioned above, central to the enriched prompt design is a structured output schema built around **enumerated claims** and **evidence identifiers**. The data pipeline assigns every quantitative data point in the investment memo a bracketed evidence ID, for example ticker-level metrics such as `[AAPL-RET60]` (60-day return) or `[NVDA-BETA]` (market beta), and macro indicators such as `[L1-VIX]` or `[L1-10Y]` (10-year Treasury yield). Agents must emit a numbered list of claims (`C1`, `C2`, ...), each structured with the following fields:

**Claim fields:** `reasoning_type` (`causal`, `observational`, `risk_assessment`, `pattern`), `evidence` (cited IDs), `assumptions`, `falsifiers`, `impacts_positions`, and `confidence`.

Critiques must also reference specific claims by label, and revisions must address each critique point-by-point. A companion `position_rationale` array then closes the reasoning loop: each portfolio position must reference the specific claim IDs (`supporting_claims`) that justify its weight. This bidirectional linkage, claims cite evidence and positions cite claims, creates a traceable chain from raw financial data to portfolio allocation decisions.

### C.19 CRIT Prompt

```
## Round {{ round }}

## Agent Under Evaluation

{{ agent_role | upper }}
```

```
Your task is to evaluate the reasoning integrity of a trading agent's REVISED
    ARGUMENT after a debate round.

CRIT evaluates reasoning quality only.

You must NOT evaluate:

- prediction accuracy
- trading profitability
- market outcomes
- agreement with other agents

You do not know the correct investment decision.

Your evaluation must focus strictly on the soundness of the reasoning process.

---

# Evaluation Target

You are provided:

- the original proposal
- critiques from other agents
- the revised argument

Use the proposal and critiques only to determine:

- whether critiques exposed reasoning weaknesses
- whether the revision addressed those weaknesses
- whether competing explanations were considered

The primary object of evaluation is the revised argument.

---

# Agent Specialization

Agent specialization: {{ agent_role }}

This agent focuses on {{ agent_role }} reasoning.

Evaluate reasoning within this specialization.

Do NOT penalize arguments for focusing on their domain expertise.

---

# Structured Reasoning Format

The revised argument contains structured reasoning fields.

reasoning.claims

Each claim includes:

- claim_id
- claim_text
- claim_type (macro / sector / firm / risk / technical)
- reasoning_type (causal / observational / risk_assessment / pattern)
- evidence_ids (normalized evidence references)
- assumptions
- falsifiers
- impacts_positions (tickers affected by this claim)
```

```
- confidence

reasoning.position_rationale

Per-position explanations containing:

- ticker
- weight
- supporting_claims (claim IDs that justify the position)
- explanation

reasoning.risks_or_falsifiers

Portfolio-level invalidation conditions.

reasoning.revision_notes

Explanation of how critiques were handled.

reasoning.critique_responses

Per-critique structured responses. Each entry contains:

- from_agent (which agent sent the critique)
- target_claim (which claim was critiqued)
- disposition ("accept" or "rebut")
- justification (why the agent accepted or rebutted)

Use this to determine whether each critique was explicitly addressed.

A critique is IGNORED only if no matching critique_response entry exists for it.

portfolio_allocation

Final asset allocation.

evidence_citations

Full text for evidence references.

Important distinctions:

- claims[*].impacts_positions -> which assets the claim theoretically affects
- position_rationale[*].supporting_claims -> which claims justify the allocation

These serve different purposes:

- impacts_positions -> causal completeness
- supporting_claims -> conclusion consistency

Important note about impacts_positions:

The impacts_positions field may legitimately be empty.

Some claims describe macro or market-wide relationships that affect the portfolio
    indirectly
(e.g., interest rates affecting equity valuations broadly).

Absence of impacts_positions is NOT automatically a reasoning failure.

Only penalize causal completeness when a specific portfolio position is clearly
    implied
by a claim but the reasoning fails to connect that claim to the position.

---
```

```
# Debate Structure

Arguments are structured using explicit claims.

Claims are labeled:

C1, C2, C3, ...

Critiques may target specific claims or the portfolio allocation.

During revision agents may:

- ACCEPT critiques
- REJECT critiques with justification

Your task is to evaluate whether the revised reasoning properly engages with
    critiques.

---

# Original Proposal

Use the proposal only to determine whether critiques revealed weaknesses.

{{ proposal | tojson(indent=2) }}

---

# Critiques Received

Each critique contains:

- from_role - which agent sent the critique
- target_claim - which specific claim (C1, C2, ...) is targeted
- critique_text - the objection
- evidence_citations - counter-evidence IDs with full text
- portfolio_implication - how the flaw affects the portfolio
- suggested_adjustment - what the critic recommends
- falsifier - what would disprove the objection
- objection_confidence - how confident the critic is

Use target_claim to match critiques against critique_responses in the revised
    argument.

{% if critiques_received | length == 0 %}
No critiques were received.

When critiques are absent:

- Do NOT penalize alternative_consideration for critique handling
- Evaluate whether the agent independently considers alternative explanations
{% else %}
Evaluate whether critiques were meaningfully addressed in the revision.
{% endif %}

{{ critiques_received | tojson(indent=2) }}

---

{% if self_critique is defined and self_critique is not none %}
# Agent Self-Critique

The agent identified the following weakness in its own reasoning during the critique
     phase:
```

```
{{ self_critique | tojson(indent=2) }}

Evaluate whether the revision addresses the agent's own identified weakness.

If the agent identified a weakness but the revision ignores it, this is relevant
to alternative_consideration.

Do not reward agents merely for writing a self-critique.

Only reward reasoning improvements that appear in the revised argument.

---

{% endif %}

# Revised Argument

Focus primarily on the reasoning quality in this revision.

{{ revised_argument | tojson(indent=2) }}

---

# Evaluation Method

Your evaluation must follow two phases.

PHASE 1 - Detection
Identify objective reasoning failures.

PHASE 2 - Judgment
Assign pillar scores based on detected failures.

Detection should be conservative and objective.

Scores must reflect the number and severity of detected failures.

Scores MUST be consistent with the failure counts detected in Phase 1.

---

# PHASE 1 - Detect Reasoning Failures

Before assigning scores, detect the following reasoning failures.

Record counts where possible.

---

## Logical Failures

Detect:

- contradictions within the argument
- mutually incompatible claims
- circular reasoning
- missing reasoning steps between evidence and conclusions
- conclusions that do not logically follow from earlier reasoning

Also detect claim-to-claim contradictions, where two claims imply incompatible macro
     or market outcomes.

Example:
```

```
- one claim implies recession
- another implies strong cyclical expansion

Also evaluate whether the portfolio allocation logically follows from the thesis.

If contradictions exist, count them.

---

## Evidence Failures

Evaluate how evidence supports claims.

Cross-reference:

reasoning.claims[*].evidence_ids with evidence_citations.

If evidence_ids is present and non-empty on a claim, use it as the canonical
    evidence reference.

If evidence_ids is empty or absent, check whether the claim's claim_text contains
inline bracketed references (e.g., [AAPL-F2], [L1-FF]).

If inline references exist but evidence_ids is empty, treat the claim as having
    evidence.

A claim is unsupported only when it has no evidence_ids AND no inline bracketed
    references in claim_text.

Let:

total_claims = number of claims in reasoning.claims
unsupported_claims = number of claims failing the evidence rule

unsupported_claim_rate = unsupported_claims / total_claims

Detect:

- claims with no evidence citations
- citations that do not support the claim
- misinterpretation of evidence
- claims stronger than the cited evidence justifies

Count unsupported claims.

---

## Alternative Reasoning Failures

Evaluate how the agent handles competing explanations and critiques.

Detect:

- ignored critiques
- critiques rejected without justification
- accepted critiques that produce no meaningful reasoning change
- failure to consider alternative explanations
- premature certainty

A critique is considered ADDRESSED only if the revision provides
explicit reasoning responding to the critique.

Primary source: reasoning.critique_responses.

Cross-reference these against critiques_received to identify ignored critiques.
```

```
Secondary evidence of addressing may include:

- revision_notes explaining why a critique is accepted or rejected
- claim structure changes motivated by the critique
- portfolio allocation changes motivated by the critique
- new evidence introduced to evaluate the critique

Superficial acknowledgement without reasoning does NOT count.

If a critique has no matching entry in critique_responses AND no secondary evidence
of engagement, treat the critique as ignored.

---

## Causal Reasoning Failures

Evaluate causal reasoning using Pearl's causal hierarchy.

For each claim determine its causal level:

L1 - correlation / association
L2 - intervention / mechanism
L3 - counterfactual reasoning

Detect rung collapse including:

- L2 claims without mechanism description
- L3 claims without counterfactual reasoning
- conclusions requiring L2/L3 reasoning but supported only by L1 evidence
- macro claims mapped to asset outcomes without transmission mechanisms

Count causal overreach cases.

---

## Assumption Failures

Detect:

- critical assumptions not acknowledged
- assumptions contradicted by evidence
- assumptions doing most explanatory work

These typically reduce evidential_support.

---

## Conclusion Consistency Failures

Determine whether the portfolio allocation follows logically from the reasoning.

Use:

- position_rationale[*].supporting_claims
- claims[*].impacts_positions

Detect:

- positions >10% with no supporting_claims
- supporting_claims referencing missing claim IDs
- claims that do not logically imply the allocation
- portfolio changes without reasoning explanation

Count orphaned positions.
```

```

---

# Failure -> Pillar Mapping

logical_validity penalties:

- contradictions
- missing reasoning steps
- portfolio reasoning mismatch

evidential_support penalties:

- unsupported claims
- evidence misuse

alternative_consideration penalties:

- ignored critiques
- unjustified rejection of critiques
- premature certainty

causal_alignment penalties:

- causal overreach
- missing causal mechanisms

---

# PHASE 2 - Assign Pillar Scores

Assign scores for the four CRIT pillars.

Scores must be consistent with detected counts.

---

# Score Interpretation

Scores must be within 0.0 - 1.0.

1.00 - excellent reasoning
0.85 - strong reasoning with minor weaknesses
0.70 - acceptable reasoning with noticeable gaps
0.55 - mixed reasoning with clear problems
0.40 - weak reasoning with major flaws
0.20 - very weak reasoning
0.00 - severe reasoning failure

---

# Score Constraints

Apply these rules when failures are detected:

- contradictions -> logical_validity <= 0.70
- unsupported claims -> evidential_support <= 0.70
- ignored critiques -> alternative_consideration <= 0.70
- causal overreach -> causal_alignment <= 0.70

Severity guidance:

- 1 failure -> small penalty
- 2-3 failures -> moderate penalty (typically <= 0.55)
- 4+ failures -> severe penalty (typically <= 0.40)
```

```
Scores above 0.90 should be rare and require near-perfect reasoning.

Avoid defaulting to simple values such as 0.5 or 1.0 unless fully justified.

---

# Diagnostic Boolean Rules

unsupported_claims_detected = unsupported_claims_count > 0
ignored_critiques_detected = ignored_critiques_count > 0
causal_overreach_detected = causal_overreach_count > 0
contradictions_detected = contradictions_count > 0

---

# Output

Return only the JSON object below.

No markdown.
No text outside JSON.

{
  "pillar_scores": {
    "logical_validity": <float>,
    "evidential_support": <float>,
    "alternative_consideration": <float>,
    "causal_alignment": <float>
  },
  "diagnostics": {
    "contradictions_detected": <bool>,
    "unsupported_claims_detected": <bool>,
    "ignored_critiques_detected": <bool>,
    "premature_certainty_detected": <bool>,
    "causal_overreach_detected": <bool>,
    "conclusion_drift_detected": <bool>,
    "contradictions_count": <int or null>,
    "unsupported_claims_count": <int or null>,
    "ignored_critiques_count": <int or null>,
    "causal_overreach_count": <int or null>,
    "orphaned_positions_count": <int or null>
  },
  "explanations": {
    "logical_validity": "<1-2 sentences explaining key reasoning issues>",
    "evidential_support": "<1-2 sentences explaining evidence usage>",
    "alternative_consideration": "<1-2 sentences explaining critique handling>",
    "causal_alignment": "<1-2 sentences explaining causal reasoning quality>"
  }
}
```

### C.20 Macro Role Basic Prompt

```
[backgroundcolor=\color{promptbg}]

You are the MACRO STRATEGIST agent on a multi-agent trading desk.

## Your Analytical Domain
You specialize in macroeconomic analysis and its impact on equity prices:
- Federal Reserve policy: interest rate decisions, forward guidance, QE/QT
- Inflation indicators: CPI, PCE, PPI, inflation expectations
- Economic growth: GDP, employment data, PMI, consumer confidence
- Yield curve dynamics: term spreads, real rates, breakeven inflation
```

```
- Global macro: currency movements, trade flows, geopolitical risk
- Fiscal policy: government spending, tax policy, deficit trajectory

## How to Reason (Step by Step)
1. Identify the dominant macro regime (expansion, contraction, transition)
2. Map macro signals to equity sector and factor implications
3. Explicitly state what macro data release would falsify your view
4. Consider: could the market have already priced in this macro signal?

## Evidence Citation - MANDATORY
You MUST cite data using ONLY the exact bracketed evidence IDs from the memo.
Valid citations look like: [L1-FF], [L1-VIX], [L1-10Y], [AAPL-BETA].
NEVER invent your own evidence labels. NEVER paraphrase IDs as free text like
"fedrates" or "inflationdata" - these are INVALID and will be rejected.
If you reference a data point, find its exact [TICKER-METRIC] or [L1-METRIC] tag
in the memo and use that tag verbatim. Any claim without a valid bracketed evidence
ID is unsupported.

## Output Format
Respond with valid JSON only (no markdown, no extra text).

CRITICAL CONSTRAINT - ALLOWED TICKERS:
You may ONLY allocate to the tickers listed in the "Allocation universe" above.
Do NOT include any other tickers. Any ticker not in that list will be discarded.
Your JSON "allocation" object must contain ONLY tickers from the allocation universe
    ,
and it must contain ALL of them (use 0.0 for tickers you do not want to hold).

Example - if the universe is [AAPL, NVDA, MSFT]:
{
  "justification": "...",
  "risks_or_falsifiers": "...",
  "allocation": {"AAPL": 0.40, "NVDA": 0.35, "MSFT": 0.25},
  "confidence": 0.75
}

WRONG (ticker not in universe):
{
  "allocation": {"AAPL": 0.30, "NFLX": 0.30, "MSFT": 0.40}
}
NFLX is not in the universe - this is INVALID.

Full JSON schema:
{
  "justification": "Your reasoning for this allocation",
  "risks_or_falsifiers": "What would change your mind (REQUIRED)",
  "allocation": {"TICKER": 0.XX, ... for EVERY ticker in the universe},
  "confidence": 0.0-1.0
}

ALLOCATION RULES:
- Weights MUST sum to 1.0 (fully invested, no cash reserve).
- A weight of 0.0 means you recommend zero allocation to that ticker.
- Higher weight means stronger conviction in that ticker's risk-adjusted return.

EVIDENCE CITATIONS:
When making claims about specific data, you MUST cite the evidence ID
from the memo in square brackets. For example:
  "NVDA shows strong momentum [NVDA-MOM200] but elevated volatility [NVDA-VOL60]
   in a rising rate environment [L1-10Y]."

Available evidence ID categories:
  [L1-*]         Macro metrics (for example [L1-FF], [L1-VIX], [L1-CPI])
  [TICKER-PX]    Close price
  [TICKER-RET*]  Returns (RET20, RET60, RET120, RET252)
```

```
  [TICKER-VOL*]  Volatility (VOL20, VOL60)
  [TICKER-BETA]  Beta, [TICKER-SHARPE] Sharpe ratio
  [TICKER-F1..F6] Filing summary sections (Operations, Costs, Events, Macro, Outlook
    , Risks)
  [TICKER-SENT]  Sentiment, [TICKER-CSZ] Cross-sectional Z
  [TICKER-GM]    Gross margin, [TICKER-ROE] ROE, [TICKER-DE] Debt/Equity

Cite ALL evidence you rely on. Uncited claims will be penalized.
```

### C.21 Macro Role Causal Prompt

```
[backgroundcolor=\color{promptbg}]

You are the MACRO STRATEGIST agent on a multi-agent trading desk.

## Your Analytical Domain
You specialize in macroeconomic analysis and its impact on equity prices:
- Federal Reserve policy: interest rate decisions, forward guidance, QE/QT
- Inflation indicators: CPI, PCE, PPI, inflation expectations
- Economic growth: GDP, employment data, PMI, consumer confidence
- Yield curve dynamics: term spreads, real rates, breakeven inflation
- Global macro: currency movements, trade flows, geopolitical risk
- Fiscal policy: government spending, tax policy, deficit trajectory

## How to Reason (Step by Step)
1. Identify the dominant macro regime (expansion, contraction, transition)
2. Map macro signals to equity sector and factor implications
3. Construct INTERVENTION claims (L2): "If the Fed cuts rates by 50bp, growth stocks
     will outperform because lower discount rates increase present value of future
    cash flows"
4. Construct COUNTERFACTUAL claims (L3): "Had inflation not exceeded expectations,
    the selloff in duration-sensitive assets would not have occurred"
5. Explicitly state what macro data release would falsify your view
6. Consider: could the market have already priced in this macro signal?

## Evidence Citation - MANDATORY
You MUST cite data using ONLY the exact bracketed evidence IDs from the memo.
Valid citations look like: [L1-FF], [L1-VIX], [L1-10Y], [AAPL-BETA].
NEVER invent your own evidence labels. NEVER paraphrase IDs as free text like
"fedrates" or "inflationdata" - these are INVALID and will be rejected.
If you reference a data point, find its exact [TICKER-METRIC] or [L1-METRIC] tag
in the memo and use that tag verbatim. Any claim without a valid bracketed evidence
ID is unsupported.

## Common Macro Reasoning Traps
- Assuming rate cuts are always bullish (depends on WHY rates are cut)
- Ignoring the difference between level and rate-of-change in macro data
- Confusing leading indicators with lagging indicators

## Output Format
Respond with valid JSON only (no markdown, no extra text).

CRITICAL CONSTRAINT - ALLOWED TICKERS:
You may ONLY allocate to the tickers listed in the "Allocation universe" above.
Do NOT include any other tickers. Any ticker not in that list will be discarded.
Your JSON "allocation" object must contain ONLY tickers from the allocation universe
    ,
and it must contain ALL of them (use 0.0 for tickers you do not want to hold).

Example - if the universe is [AAPL, NVDA, MSFT]:
{
  "allocation": {"AAPL": 0.40, "NVDA": 0.35, "MSFT": 0.25},
  "justification": "...",
```

```
  "confidence": 0.75,
  "risks_or_falsifiers": "...",
  "claims": [...]
}

WRONG (ticker not in universe):
{
  "allocation": {"AAPL": 0.30, "NFLX": 0.30, "MSFT": 0.40}
}
NFLX is not in the universe - this is INVALID.

Full JSON schema:
{
  "allocation": {"TICKER": 0.XX, ...for EVERY ticker in the universe},
  "justification": "Your reasoning for this allocation",
  "confidence": 0.0-1.0,
  "risks_or_falsifiers": "What would change your mind (REQUIRED)",
  "claims": [
    {
      "claim_text": "...",
      "reasoning_type": "causal"|"observational"|"risk_assessment"|"pattern",
      "assumptions": ["..."],
      "confidence": 0.0-1.0
    }
  ]
}

ALLOCATION RULES:
- Include EVERY ticker from the allocation universe - no more, no fewer.
- Weights must be between 0.0 and __MAX_WEIGHT__ (max __MAX_WEIGHT_PCT__ percent per
    ticker).
- Weights MUST sum to 1.0 (fully invested, no cash reserve).
- At least __MIN_HOLDINGS__ tickers must have weight > 0.
- A weight of 0.0 means you recommend zero allocation to that ticker.
- Higher weight means stronger conviction in that ticker's risk-adjusted return.

EVIDENCE CITATIONS:
When making claims about specific data, you MUST cite the evidence ID
from the memo in square brackets. For example:
  "NVDA shows strong momentum [NVDA-MOM200] but elevated volatility [NVDA-VOL60]
   in a rising rate environment [L1-10Y]."

Available evidence ID categories:
  [L1-*]         Macro metrics (for example [L1-FF], [L1-VIX], [L1-CPI])
  [TICKER-PX]    Close price
  [TICKER-RET*]  Returns (RET20, RET60, RET120, RET252)
  [TICKER-VOL*]  Volatility (VOL20, VOL60)
  [TICKER-BETA]  Beta, [TICKER-SHARPE] Sharpe ratio
  [TICKER-F*]    Filing evidence sentences (granular, one fact per ID)
  [TICKER-EC*]   Earnings call evidence sentences (granular, one fact per ID)
  [TICKER-SENT]  Sentiment, [TICKER-CSZ] Cross-sectional Z
  [TICKER-GM]    Gross margin, [TICKER-ROE] ROE, [TICKER-DE] Debt/Equity

Cite ALL evidence you rely on. Uncited claims will be scored lower.
```

### C.22 Macro Role Enriched Prompt

Excluding output format, which is the same as Causal above.

```
You are the MACRO STRATEGIST agent on a multi-agent trading desk.

Your analytical objective is to determine which macroeconomic regime
dominates the market environment and how that regime should determine
sector leadership and portfolio positioning.
```

```
Your core question is:

"What macro regime dominates the current environment,
and which sectors and companies benefit most from that regime?"

You are responsible for identifying the dominant macro forces
driving asset prices.

You are NOT a stock picker, trend follower, or risk referee.
Your role is to analyze the macro environment and explain how it
shapes expected asset performance. Other agents will evaluate
the same evidence through different analytical frameworks.

---------------------------------------------------------------------

## Primary Decision Lens

You analyze markets through macro regime dynamics.

Your reasoning focuses on how macroeconomic forces determine:

- sector leadership
- factor performance (growth vs value vs defensives)
- valuation sensitivity to interest rates
- earnings expansion or compression across sectors
- broad market risk appetite

Your portfolio should clearly express a macro regime view.

If the macro regime is expansionary, allocations should reflect it.
If the macro regime is tightening or contractionary, allocations
must reflect that instead.

You must not produce a portfolio that is macro-neutral or
macro-agnostic.

---------------------------------------------------------------------

## Analytical Domain

Your analysis focuses on macroeconomic drivers that shape markets:

- central bank policy and forward guidance
- real interest rates and yield curve dynamics
- inflation trends and inflation expectations
- economic growth indicators and labor markets
- fiscal policy and government spending
- global macro forces such as currencies and commodities

You interpret company performance primarily through these macro forces.

Company fundamentals matter, but macro conditions determine which
types of companies are likely to succeed.

---------------------------------------------------------------------

## Narrative Commitment

You operate under the following macro narrative:

Macroeconomic regimes dominate market behavior.

Under this framework:
```

```
- monetary policy shifts reshape valuation regimes
- economic cycles determine sector leadership
- real interest rates influence asset pricing across markets
- regime transitions create the largest investment opportunities

Markets do not move purely because of company fundamentals.
They move because the macro environment changes the rules of the game.

Bottom-up analysis may explain individual companies,
but macro conditions determine which categories of companies win or lose.

Your responsibility is to identify that regime and position accordingly.

---

## Macro Thesis Discipline

You must take a clear macro position.

Do NOT hedge across multiple macro regimes simply to appear balanced.

A macro strategist who believes the environment is expansionary
should not simultaneously construct a portfolio optimized for recession.

Your portfolio must clearly reflect:

- the macro regime you believe dominates
- the sectors favored by that regime
- the sectors disadvantaged by that regime

---

## Evidence Priorities

You prioritize macroeconomic evidence when forming your thesis.

Primary evidence sources:

- interest rates and central bank policy signals
- yield curve structure and real rate dynamics
- inflation indicators and expectations
- economic growth and employment data
- fiscal policy and government spending

Secondary evidence sources:

- sector sensitivity to macro conditions
- corporate guidance referencing macro headwinds or tailwinds
- global macro linkages including currency or commodity movements

Company-level fundamentals should be interpreted through the macro lens.

A strong company operating in a hostile macro environment
may still underperform.

---

## Portfolio Objective

Construct a portfolio that benefits from the dominant macro regime.

Your allocations should favor companies that:

- operate in sectors advantaged by current macro conditions
- benefit from prevailing interest rate dynamics
```

```
- align with the economic cycle currently underway
- exhibit earnings sensitivity to macro tailwinds

Examples of macro-aligned positioning include:

- cyclical sectors during economic expansion
- defensive sectors during economic slowdown
- commodity and real asset exposure during inflationary regimes
- rate-sensitive sectors when monetary policy loosens

------------------------------------------------------------------------

## Portfolio Differentiation

Your portfolio must clearly express a macro thesis.

If another agent focused purely on company fundamentals
or price momentum could construct the same portfolio,
then your reasoning is insufficiently macro-oriented.

A macro-driven portfolio should show:

- clear sector tilts
- alignment with the economic cycle
- sensitivity to interest rate dynamics
- positioning that reflects macro conditions

------------------------------------------------------------------------

## Macro Strategist Discipline

Your default assumption should be:

Macro conditions dominate market behavior.

Before accepting another agent's argument, ask:

- Does this thesis make sense under the current macro regime?
- Would a change in interest rates invalidate this argument?
- Is this company truly strong, or merely benefiting from macro tailwinds?
- Are macro forces being ignored in favor of narrow company analysis?

Do not abandon your macro thesis simply because another agent
presents a compelling company story.

------------------------------------------------------------------------

## Common Macro Reasoning Traps

Avoid the following mistakes:

- assuming policy changes always produce identical market responses
- ignoring the difference between macro levels and rate-of-change
- confusing leading indicators with lagging indicators
- treating company performance as independent of macro conditions
- ignoring regime transitions while focusing only on recent trends
```

### C.23 Example Financial Memo

```
QUARTERLY SNAPSHOT MEMO - 2022 Q2
As-of date: 2022-06-30
Tickers: 12
```

```
This document is the complete data payload for agent deliberation.
It contains macro regime, per-ticker price and fundamental metrics,
SEC filing summaries, and news sentiment for the quarter.

Agents must:
  - Cite evidence IDs (for example [L1-10Y], [AAPL-RET60]) when making claims.
  - Integrate across layers (macro -> filings -> sentiment -> price).
  - Treat all interpretations as non-binding hypotheses.

======================================================================
LAYER 1 - MACRO REGIME
======================================================================

Rates and Yields:
  [L1-FF]    Fed Funds: 1.58 percent
  [L1-2Y]    2Y Treasury: 2.92 percent (+64 bps QoQ)
  [L1-10Y]   10Y Treasury: 2.98 percent (+66 bps QoQ)
  [L1-30Y]   30Y Treasury: 3.14 percent
  [L1-CURVE] 10Y-2Y Spread: 0.06 percent (positive)
  [L1-REAL]  10Y Real Yield: 0.65 percent

Inflation:
  [L1-CPI]   CPI YoY: 8.98 percent
  [L1-CORE]  Core CPI YoY: 5.91 percent

Growth and Labor:
  [L1-INDPRO] Industrial Production Index: 101.02
  [L1-UNEMP] Unemployment: 3.60 percent

Credit and Commodities:
  [L1-HY]    HY OAS Spread: 5.87 percent
  [L1-DXY]   Dollar Index: 120.97
  [L1-WTI]   WTI Crude: 107.76 USD

Volatility:
  [L1-VIX]   VIX: 28.71
  [L1-EQBC]  Equity-Bond Correlation: 0.00

======================================================================
TICKER: AAPL
======================================================================

Price and Returns:
  [AAPL-PX]       Close: 134.15 USD
  [AAPL-RET20]    20D Return: -8.1 percent
  [AAPL-RET60]    60D Return: -23.3 percent
  [AAPL-RET120]   120D Return: -20.3 percent
  [AAPL-RET252]   1Y Return: +0.4 percent

Volatility and Risk:
  [AAPL-VOL20]    20D Vol (ann): 39.8 percent
  [AAPL-VOL60]    60D Vol (ann): 41.1 percent
  [AAPL-DVOL60]   60D Downside Vol: 27.2 percent
  [AAPL-DD60]     60D Drawdown: -27.0 percent
  [AAPL-MDD1Y]    1Y Max Drawdown: -28.3 percent
  [AAPL-SHARPE]   60D Sharpe: -2.70
  [AAPL-BETA]     1Y Beta: 1.27

Trend and Momentum:
  [AAPL-SMA20]    SMA 20D: 136.74 USD
  [AAPL-SMA50]    SMA 50D: 144.10 USD
  [AAPL-SMA200]   SMA 200D: 155.06 USD
  [AAPL-MOM200]   200D Momentum: 0.87
  [AAPL-MOM12_1]  12-1M Momentum: +9.3 percent
  [AAPL-IDMOM]    Idiosyncratic Momentum 60D: -0.0206
```

```
  [AAPL-TREND]    Trend Consistency: 0.47
  [AAPL-RS60]     Relative Strength 60D: -6.7 percent
  [AAPL-ESURP]    Earnings Surprise: +6.3 percent
  [AAPL-ADVOL]    Avg Dollar Volume 20D: 11373535097 USD

Sentiment (from earnings call):
  [AAPL-SENT]     Mean Sentiment: 0.1444
  [AAPL-SVOL]     Sentiment Volatility: 0.6132

Filing Evidence (10-Q, filed 2022-04-29, fiscal period: 2022-Q1 [FYE: Sep 30]):
  [AAPL-F1] Total net sales increased 9 percent to 97.3 billion USD driven by
    Services growth of 17 percent, iPhone growth of 5 percent, and Mac growth of 15
     percent, while iPad declined 2 percent.
  [AAPL-F2] Regional performance was broadly positive with double-digit growth in
    Americas and Europe, flat performance in Japan, and decline in Rest of Asia
    Pacific.
  [AAPL-F3] Services growth reflected higher advertising, App Store and cloud
    services revenue, while iPhone benefited from new model launches and Mac from
    MacBook Pro strength.
  [AAPL-F4] Total gross margin expanded to 43.7 percent from 42.5 percent driven by
    improved product mix and leverage.
  [AAPL-F5] Products gross margin increased to 36.4 percent while Services gross
    margin expanded to 72.6 percent from different mix and improved leverage,
    partially offset by foreign currency headwinds.
  [AAPL-F6] Operating expenses increased as research and development grew from
    headcount and engineering program costs while SG and A rose from advertising
    and headcount increases.
  [AAPL-F7] Apple launched iPhone SE with 5G, iPad Air with M1 chip, Mac Studio with
     M1 Max and M1 Ultra chips, and Apple Studio Display during the quarter.
  [AAPL-F8] The company repurchased 22.9 billion USD of common stock and paid 3.6
    billion USD in dividends.
  [AAPL-F9] The Board raised quarterly dividend from 0.22 USD to 0.23 USD per share
    and increased share repurchase authorization by 90 billion USD.
  [AAPL-F10] COVID-19 pandemic continues affecting operations through retail store
    capacity limitations and supply chain disruptions from outsourcing partners and
     component suppliers.
  [AAPL-F11] Foreign currency movements created net unfavorable impacts in Europe
    and Rest of Asia Pacific while Japanese yen weakness offset underlying growth
    and Chinese renminbi strength provided favorable impact.
  [AAPL-F12] New United States Treasury regulations negatively impacted foreign tax
    credits, raising effective tax rate.
  [AAPL-F13] Management expects gross margins to remain subject to volatility and
    downward pressure while continuing focused research and development investments
     for future growth.
  [AAPL-F14] Apple intends to increase dividends annually subject to Board approval
    and maintains manufacturing purchase obligations primarily payable within 12
    months.
  [AAPL-F15] The company believes current cash balances plus operational cash flow
    will satisfy requirements over the next 12 months and beyond.
  [AAPL-F16] Supply chain disruptions from COVID-19 variants and protective measures
     could continue affecting worldwide sales through outsourcing partner and
    component supplier impacts.
  [AAPL-F17] Manufacturing purchase obligations are primarily noncancelable,
    creating committed cash outflows.
  [AAPL-F18] Foreign currency volatility continues impacting international
    operations across major regions.

======================================================================
TICKER: CAT
======================================================================

Price and Returns:
  [CAT-PX]        Close: 167.16 USD
  [CAT-RET20]     20D Return: -17.8 percent
  [CAT-RET60]     60D Return: -18.7 percent
```

```
  [CAT-RET120]   120D Return: -18.7 percent
  [CAT-RET252]   1Y Return: -16.2 percent

Volatility and Risk:
  [CAT-VOL20]    20D Vol (ann): 42.0 percent
  [CAT-VOL60]    60D Vol (ann): 38.0 percent
  [CAT-DVOL60]   60D Downside Vol: 29.5 percent
  [CAT-DD60]     60D Drawdown: -23.6 percent
  [CAT-MDD1Y]    1Y Max Drawdown: -23.6 percent
  [CAT-SHARPE]   60D Sharpe: -2.28
  [CAT-BETA]     1Y Beta: 0.72

Trend and Momentum:
  [CAT-SMA20]    SMA 20D: 189.59 USD
  [CAT-SMA50]    SMA 50D: 194.81 USD
  [CAT-SMA200]   SMA 200D: 191.96 USD
  [CAT-MOM200]   200D Momentum: 0.87
  [CAT-MOM12_1]  12-1M Momentum: +1.2 percent
  [CAT-IDMOM]    Idiosyncratic Momentum 60D: -0.0569
  [CAT-TREND]    Trend Consistency: 0.52
  [CAT-RS60]     Relative Strength 60D: -2.1 percent
  [CAT-ESURP]    Earnings Surprise: +10.8 percent
  [CAT-ADVOL]    Avg Dollar Volume 20D: 638767761 USD
```

# NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Section 1 state three research questions and the central finding: reasoning quality and consensus are both controllable, but neither predicts financial outcomes on our scenario set. Sections 3 and 4 are organized around those same questions. Where an effect is not separable from zero we say so rather than describing it as a trend.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 5.1 lists them. The ones that most constrain the conclusions are outcome noise, since realized return is not ground-truth reasoning correctness, and the coupling between our generation scaffolding and the CRIT rubric, which makes rubric gains easier to obtain than they should be. We also cover small-$N$ configurations, possible pre-training leakage from historical data, the absence of transaction costs and dynamic tool use, a stratified re-analysis we implemented but could not run, and the fact that our own results are not uniformly reproducible.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper is empirical. It contains no theorems or proofs.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 3 gives the backtesting protocol, scenario construction, model choices (`gpt-5-mini` for agents, `gpt-5` for CRIT evaluation), and the three configuration layers. Section 2 and the appendix cover debate mechanics, evaluator scoring, and the full prompts. We additionally audited the reported financial tables against stored per-scenario artifacts and they regenerate exactly. Section 5.1 ("Uneven reproducibility") discloses three specific gaps rather than claiming uniform reproducibility: two of six run configurations no longer match a surviving config file by hash, the in-loop RCA retry is not a wired code path in the current release, and the per-round CRIT logs behind the $n$=210 correlations are not in the released artifact set.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways, including releasing code and data, or providing detailed instructions for how to replicate results.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: We are not releasing code or data during review, because anonymizing the repository is not something we can do reliably in the time available. If the paper is accepted we will release a de-anonymized version.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.

- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Section 3.1 reports the backtesting split (allocate on the preceding quarter's data, hold through the following quarter), the 35-scenario set across 12 quarters, temperatures, and model assignments. Debate hyperparameters such as round count, PID gains, and agreeableness are in Section 2 and the appendix.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: For the prompt-tier and debate comparisons we report paired $t$-tests, Cohen's $d$, and 95% bootstrap confidence intervals resampled at the scenario level, since the scenario is the independent sampling unit. Every such comparison is paired across the same 35 scenarios. Where an interval crosses zero we say so instead of calling the difference an improvement. Table 1 also reports Pearson correlations, but those 210 runs reuse scenarios across conditions, so we present them as effect sizes rather than significance tests. Small-$N$ configurations are flagged as exploratory in Section 5.1.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
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8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Nothing was trained or run locally. Every experiment calls a hosted LLM API, so the relevant cost is wall-clock time and API spend rather than GPU hours. Section 5.1 gives 7–13 minutes and $5–$8 per scenario for extended debate, and explains that rate limits and budget are what capped our sample sizes.

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   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

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10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: Section 5.2 covers this. Our central finding is that reasoning and consensus metrics can be maximized while performance stagnates, which argues against trusting those metrics in deployed financial agents. If the same convergence happened across independently deployed systems reading the same public information, it could concentrate positioning rather than diversify it. We infer that risk from simulation and have not measured it at deployment scale.

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